Appendix A1: Solutions for Regime-monotone Mixed MPEs when $\bar{K} = 1, 2, 3$ and Selected Proofs

Let $\sigma_k \in [0, 1]$ denote the probability that a representative agent chooses X in state k. Under a symmetric strategy profile, the probability that all other $n-1$ agents choose X is $q_k \equiv \sigma_k^{n-1}$. Thus, $\sigma_k = q_k^{1/(n-1)}$. The analysis below applies to the group sizes used in the experiment, $n \in \{2, 4\}$, and more generally to any $n \geq 2$. Let $\delta \in (0, 1)$ denote the continuation probability. Recall the stage payoffs:

$$u_{XX} = 1, \qquad u_{XY} = 0, \qquad u_Y \in (0, 1).$$

Let V_k denote the continuation value at the beginning of a round in state k, and let $Q_k(X)$ and $Q_k(Y)$ denote the corresponding action values. Choosing Y ensures that the current outcome is not all-X, so the next state is 0 and the overwrite rule applies if the shock arrives. Hence, for every k,

$$Q_k(Y) = \delta\big(u_Y + V_0\big) + (1-\delta)(-d). \tag{5}$$

If the agent chooses X, then with probability q_k all other $n-1$ agents also choose X. If an all-X outcome does not yet reach the threshold, i.e. $k \leq \bar{K} - 2$, then

$$Q_k(X) = q_k\Big[\delta\big(1 + V_{k+1}\big) + (1-\delta)(-d)\Big] + (1-q_k)\Big[\delta V_0 + (1-\delta)(-d)\Big]. \tag{6}$$

At $k = \bar{K} - 1$, an all-X outcome reaches the protected state $\bar{K}$, so

$$Q_{\bar{K}-1}(X) = q_{\bar{K}-1}\Big[\delta\big(1 + V_{\bar{K}}\big) + (1-\delta)\Big] + (1-q_{\bar{K}-1})\Big[\delta V_0 + (1-\delta)(-d)\Big]. \tag{7}$$

Finally, at the protected state $k = \bar{K}$,

$$Q_{\bar{K}}(X) = q_{\bar{K}}\Big[\delta\big(1 + V_{\bar{K}}\big) + (1-\delta)\Big] + (1-q_{\bar{K}})\Big[\delta V_0 + (1-\delta)(-d)\Big]. \tag{8}$$

In any state in which agents mix, $\sigma_k \in (0, 1)$ and hence $q_k \in (0, 1)$. The indifference condition is

$$Q_k(X) = Q_k(Y) = V_k.$$

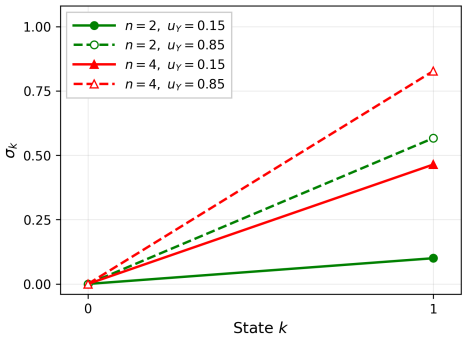
Since $q_k = \sigma_k^{n-1}$, solving for q_k is sufficient; the corresponding individual mixing probability is $\sigma_k = q_k^{1/(n-1)}$. Below is the complete illustration of all regime-monotone mixed MPEs for $\bar{K} = 1, 2, 3$. Note that the point values in the corresponding figures take the parameterization of $\delta = 0.8, d = 1$, and $n = 2$ (in green), $n = 4$ (in red), $u_Y = 0.15$ (in solid lines), $u_Y = 0.85$

(in dashed lines). Some candidates have blue tags to explain why they fail to be the final solution for the regime-monotone mixed MPE. Selected proofs of candidates without a blue tag are shown right after. Section A1.4 provides the proofs for pure MPEs, and Section A1.5 counts the regime patterns.

A1.1 Solutions when $\bar{K} = 1$

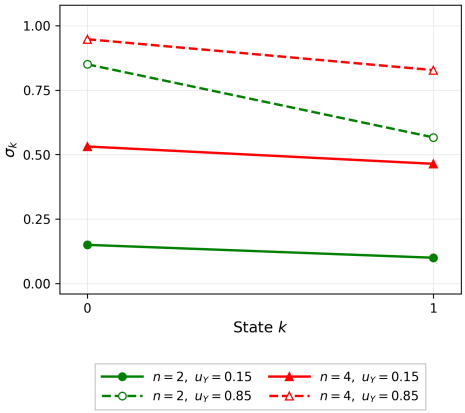
[Candidate $(0m)$**]: off-path**

$$V_0 = \frac{\delta}{1-\delta} u_Y - d, \qquad V_1 = V_0,$$
$$\sigma_0 = 0, \qquad \sigma_1 = \left(\frac{\delta u_Y}{1 + (1-\delta)d}\right)^{\frac{1}{n-1}}.$$



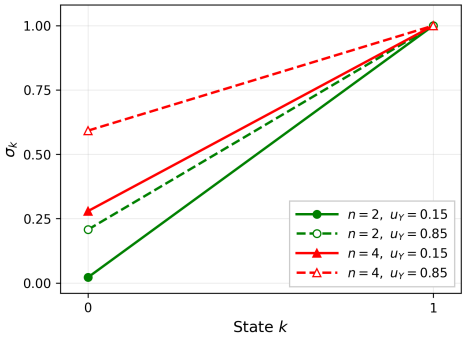
[Candidate (mm)**]: not monotone**

$$V_0 = \frac{\delta}{1-\delta} u_Y - d, \qquad V_1 = V_0$$
$$\sigma_0 = u_Y^{\frac{1}{n-1}}, \qquad \sigma_1 = \left(\frac{\delta u_Y}{1 + (1-\delta)d}\right)^{\frac{1}{n-1}}.$$



[Candidate $(m1)$**]**

$$V_0 = \frac{\delta}{1-\delta} u_Y - d, \qquad V_1 = \frac{1}{1-\delta},$$
$$\sigma_0 = \left(\frac{\delta(1-\delta)u_Y}{1 - \delta^2 u_Y + (1-\delta)d}\right)^{\frac{1}{n-1}}, \qquad \sigma_1 = 1.$$



Proof. Consider

$$\sigma_0 \in (0,1), \qquad \sigma_1 = 1, \qquad \text{or equivalently,} \qquad q_0 \in (0,1), \qquad q_1 = 1.$$

Since Y is played with positive probability at $k = 0$,

$$V_0 = Q_0(Y)$$

Using (5),

$$V_0 = \delta\big(u_Y + V_0\big) + (1-\delta)(-d),$$

which gives

$$V_0 = \frac{\delta}{1-\delta} u_Y - d. \tag{9}$$

At $k = 1$, the candidate prescribes X with probability one. The protected state is therefore absorbing along the equilibrium path:

$$V_1 = \delta\big(1 + V_1\big) + (1-\delta).$$

Hence

$$V_1 = \frac{1}{1-\delta}. \tag{10}$$

Mixing at $k = 0$ requires $Q_0(X) = Q_0(Y) = V_0$. Since an all-X outcome reaches protection immediately,

$$V_0 = q_0\Big[\delta\big(1 + V_1\big) + (1-\delta)\Big] + (1-q_0)\Big[\delta V_0 + (1-\delta)(-d)\Big].$$

Substituting (9) and (10) and rearranging yields

$$q_0 = \frac{\delta(1-\delta)u_Y}{1 - \delta^2 u_Y + (1-\delta)d}. \tag{11}$$

Therefore,

$$\sigma_0 = \left[\frac{\delta(1-\delta)u_Y}{1 - \delta^2 u_Y + (1-\delta)d}\right]^{1/(n-1)}. \tag{12}$$

Writing $q_0 = N/D$, we have

$$D - N = 1 - \delta u_Y + (1-\delta)d > 1 - u_Y > 0.$$

Thus $D > N > 0$, implying $q_0 \in (0,1)$ and hence $\sigma_0 = q_0^{1/(n-1)} \in (0,1)$. It remains to verify that X is optimal at $k = 1$. Under the candidate,

$$Q_1(X) = V_1, \qquad Q_1(Y) = V_0.$$

Thus X is optimal if $V_1 \geq V_0$, or

$$\frac{1}{1-\delta} \geq \frac{\delta}{1-\delta} u_Y - d,$$

which is equivalent to

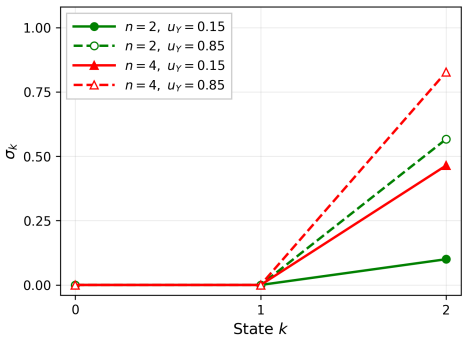
$$1 \geq \delta u_Y - (1-\delta)d.$$

This holds because $1 > u_Y > \delta u_Y$ and $d > 0$. □

A1.2 Solutions when $\bar{K} = 2$

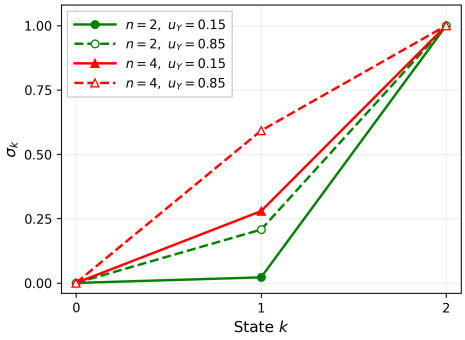
[Candidate $(00m)$]: off-path

$$V_0 = V_1 = V_2 = \frac{\delta}{1-\delta} u_Y - d,$$
$$\sigma_0 = 0, \qquad \sigma_1 = 0, \qquad \sigma_2 = \left(\frac{\delta u_Y}{1 + (1-\delta)d}\right)^{\frac{1}{n-1}}.$$



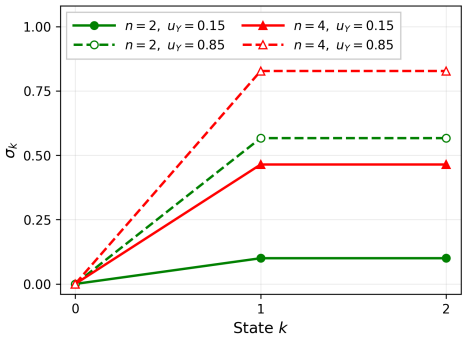
[Candidate $(0m1)$]: off-path

$$V_0 = V_1 = \frac{\delta}{1-\delta} u_Y - d, \qquad V_2 = \frac{1}{1-\delta},$$
$$\sigma_0 = 0, \qquad \sigma_1 = \left(\frac{\delta(1-\delta)u_Y}{1 - \delta^2 u_Y + (1-\delta)d}\right)^{\frac{1}{n-1}}, \qquad \sigma_2 = 1.$$



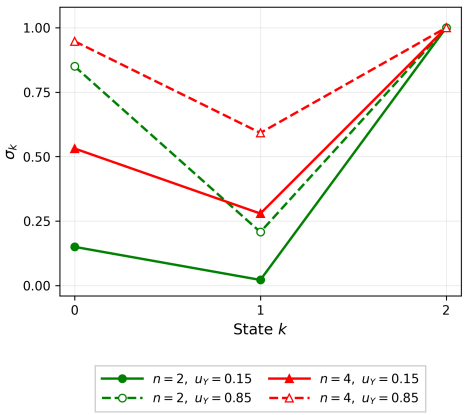
[Candidate $(0mm)$]: off-path

$$V_0 = V_1 = V_2 = \frac{\delta}{1-\delta} u_Y - d,$$
$$\sigma_0 = 0, \qquad \sigma_1 = \sigma_2 = \left(\frac{\delta u_Y}{1 + (1-\delta)d}\right)^{\frac{1}{n-1}}.$$



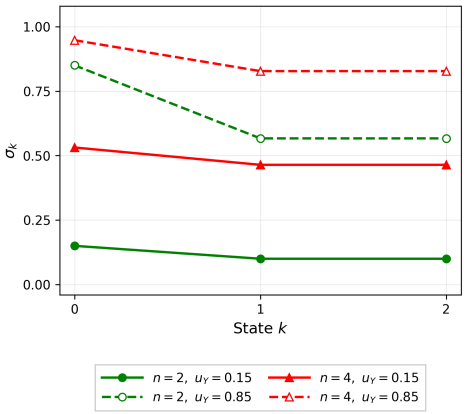
[Candidate $(mm1)$]: not monotone

$$V_0 = V_1 = \frac{\delta}{1-\delta} u_Y - d, \qquad V_2 = \frac{1}{1-\delta},$$
$$\sigma_0 = u_Y^{\frac{1}{n-1}},$$
$$\sigma_1 = \left(\frac{\delta(1-\delta)u_Y}{1-\delta^2 u_Y + (1-\delta)d} \right)^{\frac{1}{n-1}}, \qquad \sigma_2 = 1.$$



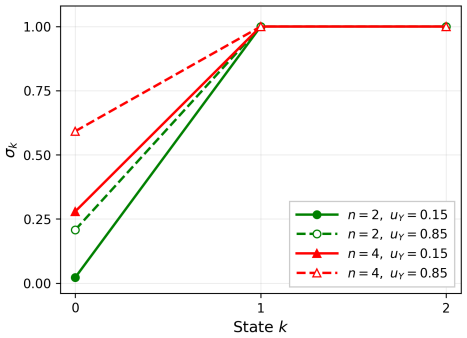
[Candidate (mmm)]: not monotone

$$V_0 = V_1 = V_2 = \frac{\delta}{1-\delta} u_Y - d,$$
$$\sigma_0 = u_Y^{\frac{1}{n-1}}, \qquad \sigma_1 = \sigma_2 = \left(\frac{\delta u_Y}{1 + (1-\delta)d} \right)^{\frac{1}{n-1}}.$$



[Candidate $(m11)$]

$$V_0 = \frac{\delta}{1-\delta} u_Y - d, \qquad V_1 = V_2 = \frac{1}{1-\delta},$$
$$\sigma_0 = \left(\frac{(1-\delta)u_Y}{(2-\delta) - \delta u_Y + (1-\delta)d} \right)^{\frac{1}{n-1}}, \quad \sigma_1 = \sigma_2 = 1.$$



Proof. Consider

$$\sigma_0 \in (0,1), \qquad \sigma_1 = \sigma_2 = 1, \qquad \text{or equivalently,} \qquad q_0 \in (0,1), \qquad q_1 = q_2 = 1.$$

At the protected state,

$$V_2 = \delta\big(1 + V_2\big) + (1-\delta),$$

so

$$V_2 = \frac{1}{1-\delta}. \tag{13}$$

At $k = 1$, choosing X reaches protection in the current round:

$$V_1 = \delta\big(1 + V_2\big) + (1 - \delta) = \frac{1}{1 - \delta}.$$

Thus,

$$V_1 = V_2 = \frac{1}{1 - \delta}. \tag{14}$$

Since agents mix at $k = 0$,

$$V_0 = Q_0(Y),$$

and therefore, as in (9),

$$V_0 = \frac{\delta}{1 - \delta} u_Y - d. \tag{15}$$

At $k = 0$, an all-X outcome moves the state to $1 < \bar{K}$, so the overwrite rule still applies if a shock arrives. The indifference condition is

$$q_0(1 + V_1) + (1 - q_0)V_0 = u_Y + V_0.$$

Hence

$$q_0 = \frac{u_Y}{1 + V_1 - V_0}.$$

Substituting (14) and (15),

$$q_0 = \frac{(1 - \delta)u_Y}{(2 - \delta) - \delta u_Y + (1 - \delta)d}. \tag{16}$$

Therefore,

$$\sigma_0 = q_0^{1/(n-1)}. \tag{17}$$

Write q_0 as $q_0 = N/D$, then

$$D - N = (2 - \delta) - u_Y + (1 - \delta)d > (2 - \delta)u_Y - u_Y > 0.$$

Thus $q_0 \in (0, 1)$ and therefore $\sigma_0 \in (0, 1)$. For $k = 1, 2$, $Q_k(Y) = V_0$ while

$$Q_1(X) = V_1, \qquad Q_2(X) = V_2.$$

Since

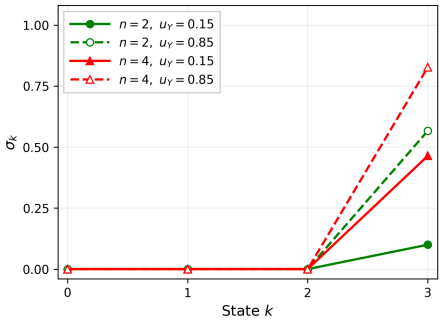
$$V_1 = V_2 = \frac{1}{1 - \delta} > \frac{\delta}{1 - \delta} u_Y - d = V_0,$$

choosing X is optimal in both states. □

A1.3 Solutions when $\bar{K} = 3$

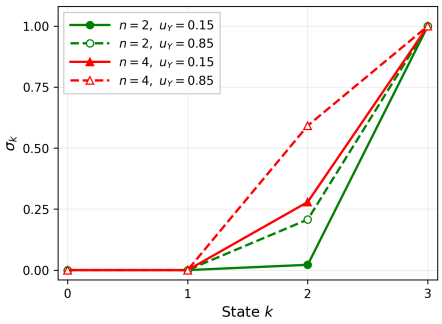
[Candidate $(000m)$]: off-path

$$V_0 = V_1 = V_2 = V_3 = \frac{\delta}{1-\delta} u_Y - d,$$
$$\sigma_0 = \sigma_1 = \sigma_2 = 0, \qquad \sigma_3 = \left(\frac{\delta u_Y}{1 + (1-\delta)d} \right)^{\frac{1}{n-1}}.$$



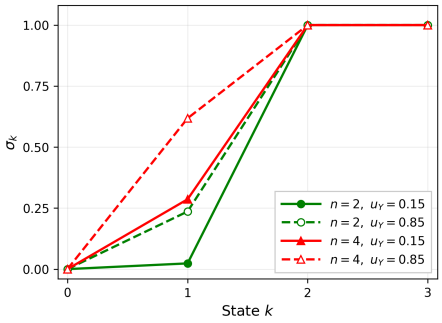
[Candidate $(00m1)$]: off-path

$$V_0 = V_1 = V_2 = \frac{\delta}{1-\delta} u_Y - d, \qquad V_3 = \frac{1}{1-\delta},$$
$$\sigma_0 = \sigma_1 = 0, \qquad \sigma_2 = \left(\frac{\delta(1-\delta) u_Y}{1 - \delta^2 u_Y + (1-\delta)d} \right)^{\frac{1}{n-1}},$$
$$\sigma_3 = 1.$$



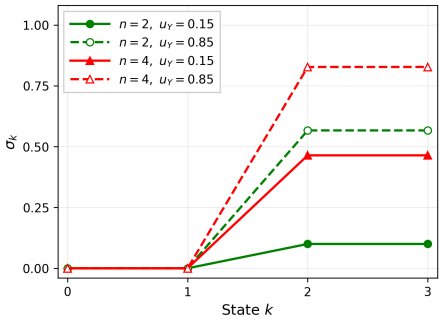
[Candidate $(0m11)$]: off-path

$$V_0 = V_1 = \frac{\delta}{1-\delta} u_Y - d, \qquad V_2 = V_3 = \frac{1}{1-\delta},$$
$$\sigma_0 = 0, \qquad \sigma_1 = \left(\frac{(1-\delta) u_Y}{(2 - \delta - \delta u_Y) + (1-\delta)d} \right)^{\frac{1}{n-1}},$$
$$\sigma_2 = \sigma_3 = 1.$$



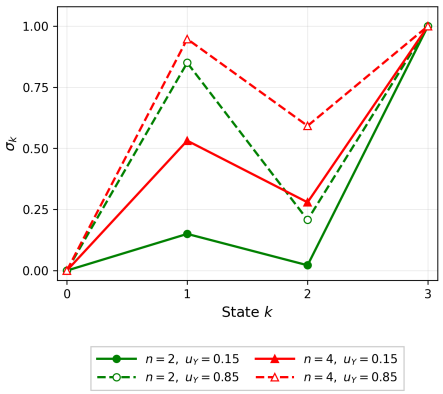
[Candidate $(00mm)$]: off-path

$$V_0 = V_1 = V_2 = V_3 = \frac{\delta}{1-\delta} u_Y - d,$$
$$\sigma_0 = \sigma_1 = 0, \qquad \sigma_2 = \sigma_3 = \left(\frac{\delta u_Y}{1 + (1-\delta)d} \right)^{\frac{1}{n-1}}.$$



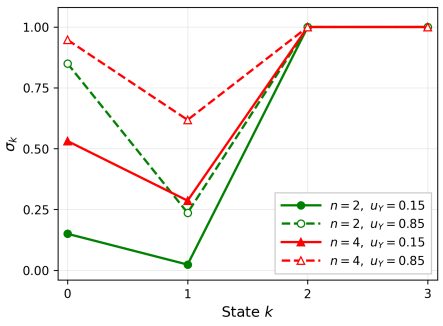
[Candidate $(0mm1)$]: off-path, not monotone

$$V_0 = V_1 = V_2 = \frac{\delta}{1-\delta} u_Y - d, \qquad V_3 = \frac{1}{1-\delta},$$
$$\sigma_0 = 0, \qquad \sigma_1 = u_Y^{\frac{1}{n-1}},$$
$$\sigma_2 = \left(\frac{\delta(1-\delta)u_Y}{1-\delta^2 u_Y + (1-\delta)d} \right)^{\frac{1}{n-1}} \qquad \sigma_3 = 1.$$



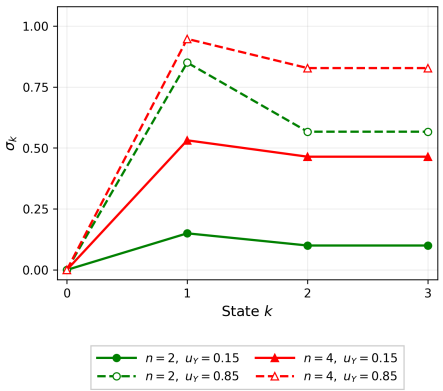
[Candidate $(mm11)$]: not monotone

$$V_0 = V_1 = \frac{\delta}{1-\delta} u_Y - d, \qquad V_2 = V_3 = \frac{1}{1-\delta},$$
$$\sigma_0 = u_Y^{\frac{1}{n-1}},$$
$$\sigma_1 = \left(\frac{(1-\delta)u_Y}{(2-\delta-\delta u_Y) + (1-\delta)d} \right)^{\frac{1}{n-1}},$$
$$\sigma_2 = \sigma_3 = 1.$$



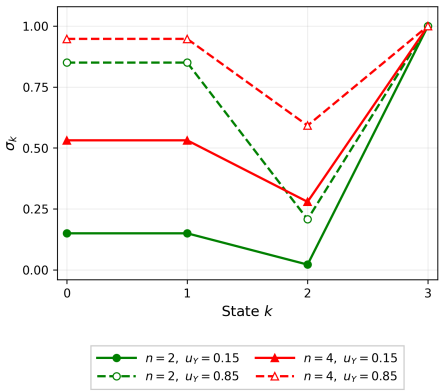
[Candidate $(0mmm)$]: off-path, not monotone

$$V_0 = V_1 = V_2 = V_3 = \frac{\delta}{1-\delta} u_Y - d,$$
$$\sigma_0 = 0, \qquad \sigma_1 = u_Y^{\frac{1}{n-1}},$$
$$\sigma_2 = \sigma_3 = \left(\frac{\delta u_Y}{1 + (1-\delta)d} \right)^{\frac{1}{n-1}}.$$



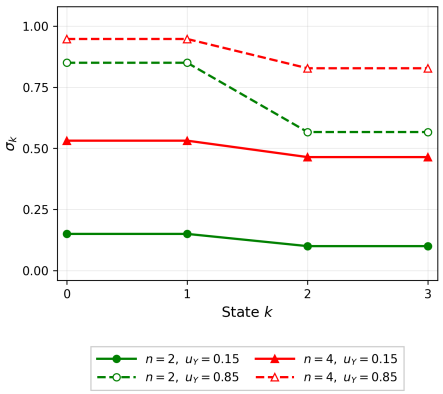
[Candidate $(mmm1)$]: not monotone

$$V_0 = V_1 = V_2 = \frac{\delta}{1-\delta} u_Y - d, \qquad V_3 = \frac{1}{1-\delta},$$
$$\sigma_0 = \sigma_1 = u_Y^{\frac{1}{n-1}},$$
$$\sigma_2 = \left(\frac{\delta(1-\delta)u_Y}{1-\delta^2 u_Y + (1-\delta)d} \right)^{\frac{1}{n-1}}, \qquad \sigma_3 = 1.$$



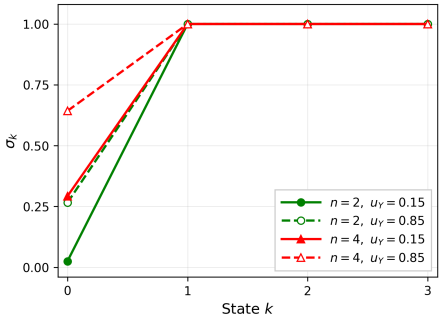
[Candidate $(mmmm)$**]: not monotone**

$$V_0 = V_1 = V_2 = V_3 = \frac{\delta}{1-\delta} u_Y - d,$$
$$\sigma_0 = \sigma_1 = u_Y^{\frac{1}{n-1}},$$
$$\sigma_2 = \sigma_3 = \left(\frac{\delta u_Y}{1+(1-\delta)d}\right)^{\frac{1}{n-1}}.$$



[Candidate $(m111)$**]**

$$V_0 = \frac{\delta}{1-\delta} u_Y - d, \qquad V_1 = \frac{\delta(2-\delta)}{1-\delta} - (1-\delta)d,$$
$$V_2 = V_3 = \frac{1}{1-\delta},$$
$$\sigma_0 = \left(\frac{(1-\delta)u_Y}{1+\delta(1-\delta)(1+d)-\delta u_Y}\right)^{\frac{1}{n-1}},$$
$$\sigma_1 = \sigma_2 = \sigma_3 = 1.$$



Proof. Consider

$$\sigma_0 \in (0,1), \qquad \sigma_1 = \sigma_2 = \sigma_3 = 1, \qquad \text{or equivalently,} \qquad q_0 \in (0,1), \qquad q_1 = q_2 = q_3 = 1.$$

At $k = 3$,

$$V_3 = \delta(1 + V_3) + (1-\delta),$$

so

$$V_3 = \frac{1}{1-\delta}. \tag{18}$$

At $k = 2$, choosing X reaches protection in the current round, implying

$$V_2 = \frac{1}{1-\delta} = V_3. \tag{19}$$

At $k = 1$, an all-X outcome moves the state to $2 < \bar{K}$, so the overwrite rule still applies if the shock arrives:

$$V_1 = \delta(1 + V_2) + (1-\delta)(-d).$$

Substituting (19),

$$V_1 = \frac{\delta(2-\delta)}{1-\delta} - (1-\delta)d. \tag{20}$$

Since agents mix at $k = 0$,

$$V_0 = \frac{\delta}{1-\delta} u_Y - d. \tag{21}$$

At $k = 0$, the indifference condition gives

$$q_0(1 + V_1) + (1 - q_0)V_0 = u_Y + V_0.$$

Therefore,

$$q_0 = \frac{u_Y}{1 + V_1 - V_0}.$$

Substituting (20) and (21),

$$q_0 = \frac{(1-\delta)u_Y}{(1 + \delta - \delta^2) - \delta u_Y + \delta(1-\delta)d}. \tag{22}$$

Hence

$$\sigma_0 = q_0^{1/(n-1)}. \tag{23}$$

Write q_0 as $q_0 = N/D$, then

$$D - N = (1 + \delta - \delta^2) - u_Y + \delta(1-\delta)d > (1 + \delta - \delta^2)u_Y - u_Y > 0.$$

Thus $q_0 \in (0, 1)$ and $\sigma_0 \in (0, 1)$. For $k = 2, 3$, $Q_k(X) = u_{XX}/(1-\delta) > V_0 = Q_k(Y)$. At $k = 1$,

$$Q_1(X) = V_1, \qquad Q_1(Y) = V_0,$$

and

$$V_1 - V_0 = \frac{\delta}{1-\delta}\Big[(2 - \delta) - u_Y\Big] + \delta d > 0.$$

Thus X is optimal at every $k \geq 1$. □

A1.4 Proofs for Pure MPEs

Lemma 1 (Best response at a given state). Fix a state k.

1. If at least one other member chooses Y at state k, then Y is a dominant best response.
2. If all other members choose X at state k, then X is a best response.

Proof. If at least one other member chooses Y, choosing X cannot produce a group outcome of all-X, so the current payoff from X is $u_{XY} = 0$, and the value of the next state k' is 0. Choosing Y yields a payoff $u_Y \in (0, 1)$ and also induces $k' = 0$. Hence Y strictly dominates

X. If all other members choose X, choosing X yields current payoff $u_{XX} = 1$ and moves the value of the next state to $k' = \min\{k+1, \bar{K}\}$, which is (weakly) closer to the threshold $\bar{K}$; while deviating to Y yields current payoff $u_Y \in (0,1)$ and resets the value of the next state to $k' = 0$. Therefore, X is a best response. □

Proposition 3 (All pure symmetric MPEs). Every symmetric pure Markov strategy profile $a(\cdot)$ with $a(k) \in \{X, Y\}$ for all k is a Markov Perfect Equilibrium. Hence there are exactly $2^{\bar{K}+1}$ pure symmetric MPEs.

Proof. Take any symmetric pure Markov strategy $a(\cdot)$. At any state k, if $a(k) = Y$, then all other members also choose Y at that state, so by Lemma 1, Y is a best response. If $a(k) = X$, then all other members also choose X, so by Lemma 1, X is a best response. Therefore, the prescribed action is optimal at every state, and $a(\cdot)$ is a pure symmetric MPE. Since at each of the $\bar{K}+1$ states the action can be chosen independently as either X or Y, the total number of pure symmetric MPEs is $2^{\bar{K}+1}$. □

Proof of Proposition 1. Let $a(\cdot)$ be a monotone pure Markov strategy. If $a(k) \equiv Y$ for all k, then this is the case $r = \bar{K}+1$. Otherwise, let $r \equiv \min\{k : a(k) = X\}$. By monotonicity, $a(k) = Y$ for all $k < r$ and $a(k) = X$ for all $k \geq r$, so $a(\cdot) = a^{(r)}(\cdot)$ for some $r \in \{0, 1, \ldots, \bar{K}\}$. The case $r = 0$ gives $a(k) \equiv X$. Conversely, each threshold strategy $a^{(r)}(\cdot)$ is a symmetric pure MPE by Proposition 3, since every symmetric pure Markov strategy is a pure MPE. Hence, the monotone pure symmetric MPEs are exactly those listed above. □

A1.5 Counting Regime Patterns

Proposition 4 (Counting mixed-MPE regime patterns). Consider symmetric, stationary Markov profiles on the state space $\{0, 1, \ldots, \bar{K}\}$. In any stationary MPE, each state k must fall into exactly one of three regimes:

$$\sigma_k = \begin{cases} 1, & \text{if } Q_k(X) > Q_k(Y), \\ 0, & \text{if } Q_k(X) < Q_k(Y), \\ \in (0,1), & \text{only if } Q_k(X) = Q_k(Y). \end{cases}$$

Hence, the number of distinct regime patterns is $3^{\bar{K}+1}$, the number of pure (no-mixing) patterns is $2^{\bar{K}+1}$, and the number of mixed *candidate* regime patterns is $3^{\bar{K}+1} - 2^{\bar{K}+1}$.

Definition 3 (Regime-monotone mixed Markov profile). A stationary mixed Markov profile is regime-monotone if its regime pattern is non-decreasing in the order $0 < m < 1$; that is, whenever $k < \bar{K}$,

$$\sigma_k = 1 \;\Rightarrow\; \sigma_{k+1} = 1, \qquad \sigma_k = m \;\Rightarrow\; \sigma_{k+1} \in \{m, 1\}.$$

Equivalently, σ_k can move only "up" in the order $0 \to m \to 1$, never down.

Proposition 5 (Counting regime-monotone mixed candidates). Fix $\bar{K} \geq 1$. Then the number of regime-monotone mixed candidate patterns equals

$$\binom{\bar{K}+3}{2} - (\bar{K}+2) = \frac{(\bar{K}+1)(\bar{K}+2)}{2}.$$

Proof. A regime-monotone pattern over $\{0, m, 1\}$ on the state space $\{0, 1, \ldots, \bar{K}\}$ must be of the form

$$\underbrace{0 \cdots 0}_{z}\ \underbrace{m \cdots m}_{r}\ \underbrace{1 \cdots 1}_{o}, \qquad z, r, o \geq 0, \quad z + r + o = \bar{K} + 1.$$

Thus, such patterns are in one-to-one correspondence with triples (z, r, o) of nonnegative integers summing to $\bar{K}+1$. By stars-and-bars there are $\binom{\bar{K}+3}{2}$ such triples. Among them, those with $r = 0$ are pure 0/1 threshold patterns, and there are $\bar{K}+2$ of these. Subtracting yields $\frac{(\bar{K}+1)(\bar{K}+2)}{2}$. □

Table A1.1: Regime-monotone and Monotone Mixed MPEs for $\bar{K} = 1, 2, 3$

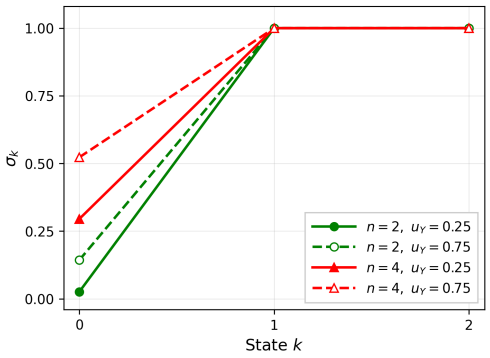
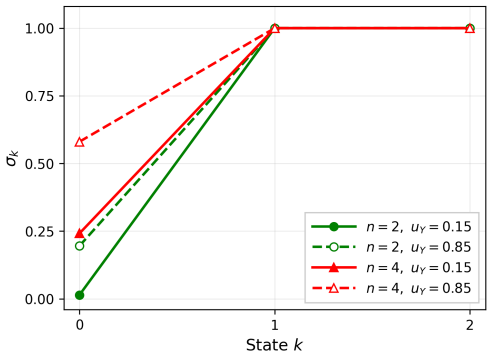
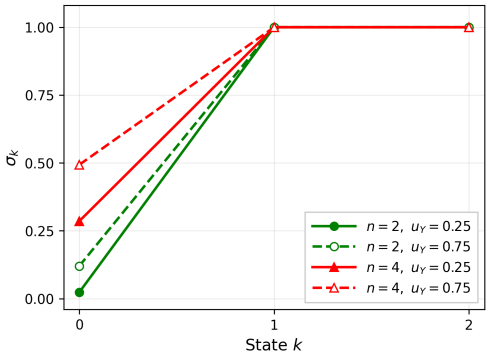
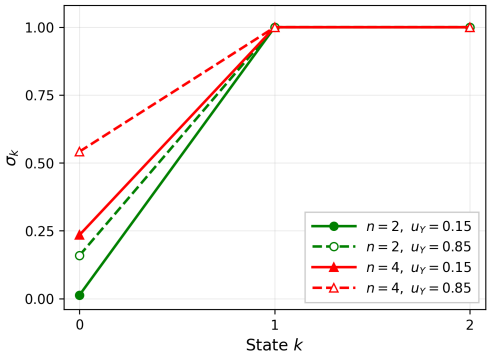
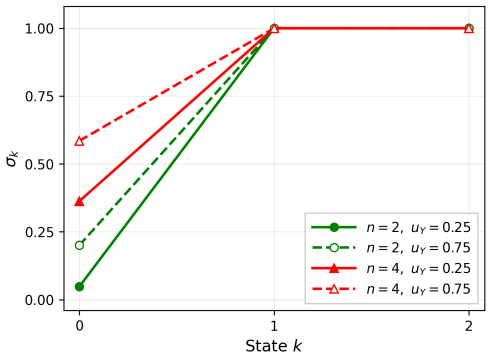
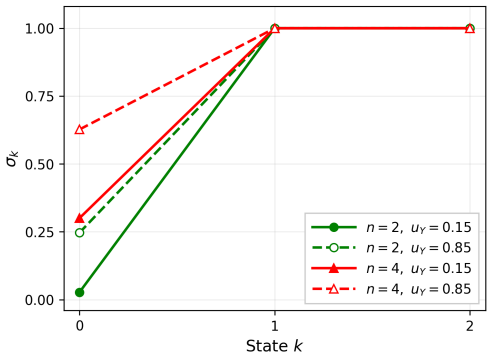
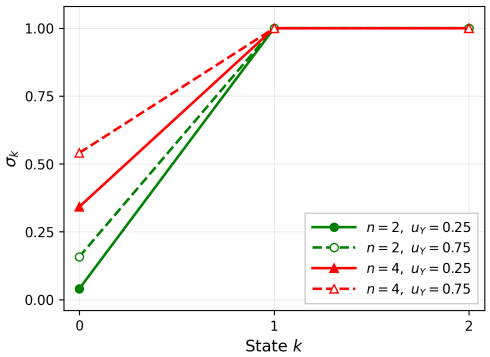
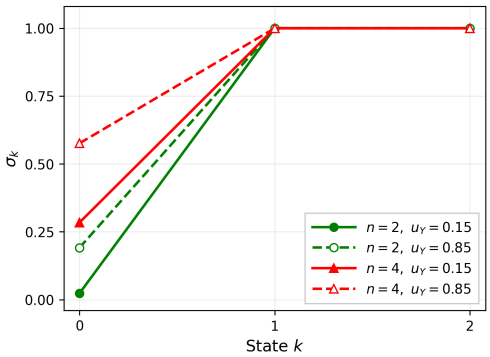
	$\bar{K}=1$		$\bar{K}=2$		$\bar{K}=3$	
	N	$\{\sigma_k\}$ profiles	N	$\{\sigma_k\}$ profiles	N	$\{\sigma_k\}$ profiles
Regime-monotone	3	$(0m)$, (mm), $(m1)$	6	$(00m)$, $(0mm)$, $(0m1)$, (mmm), $(mm1)$, $(m11)$	10	$(000m)$, $(00mm)$, $(00m1)$, $(0mmm)$, $(0mm1)$, $(0m11)$, $(mmmm)$, $(mmm1)$, $(mm11)$, $(m111)$
Monotone	2	$(0m)$, $(m1)$	4	$(00m)$, $(0mm)$, $(0m1)$, $(m11)$	5	$(000m)$, $(00mm)$, $(00m1)$, $(0m11)$, $(m111)$
Monotone (on-path)	1	$(m1)$	1	$(m11)$	1	$(m111)$

Note: The listed profiles are nested across rows: the monotone mixed MPEs form a subset of the regime-monotone candidates, and the on-path monotone mixed MPEs form a subset of the monotone mixed MPEs. m denotes interior mixing $\sigma_k \in (0, 1)$; 0 and 1 denote $\sigma_k = 0$ and $\sigma_k = 1$.

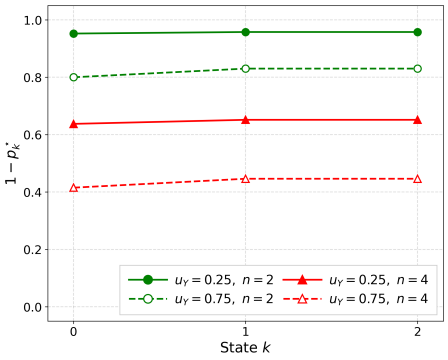
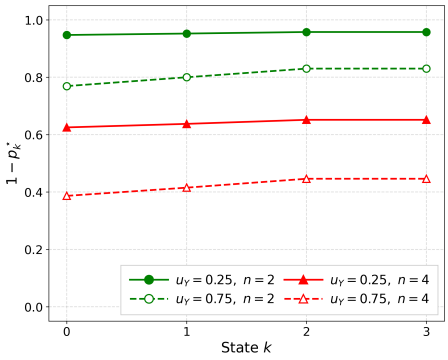
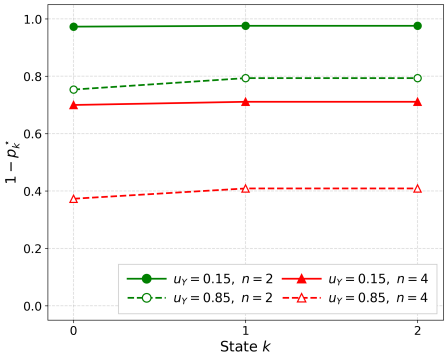
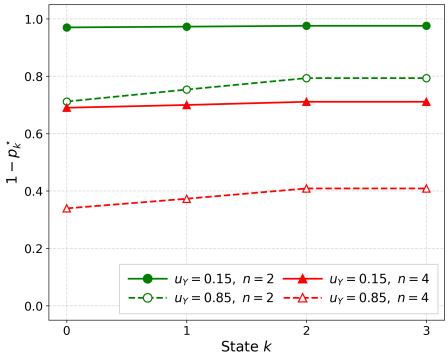
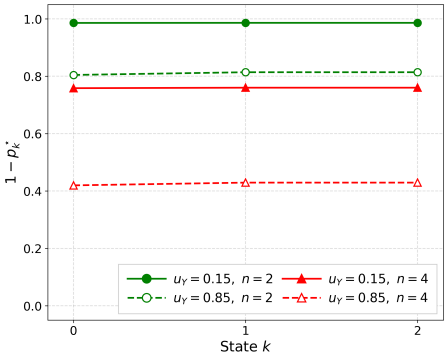
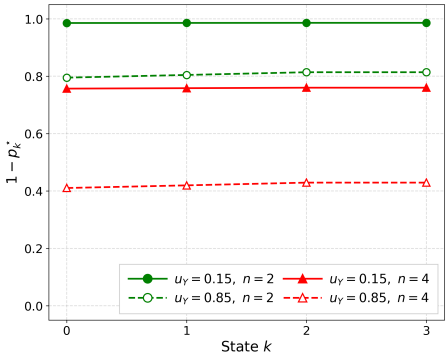
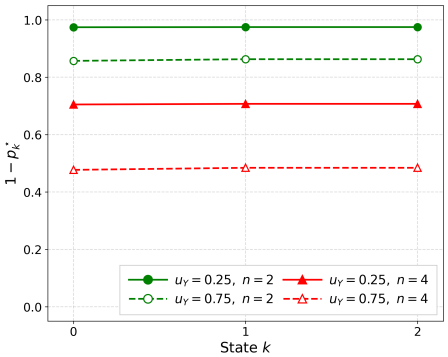
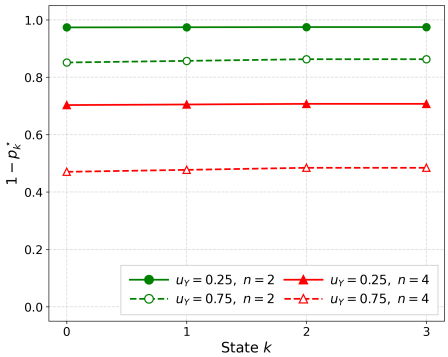
Definition 3 restricts only the potential shape of the profile. It does not guarantee that the solved mixing probabilities satisfy $\sigma_0 \leq \sigma_1 \leq \cdots \leq \sigma_{\bar{K}}$. Table A1.1 lists all regime-monotone candidates and the monotone mixed MPEs for $\bar{K} = 1, 2, 3$.

Appendix A2: Parameter Comparisons

A2.1 Parameter Comparison for the $(m11)$ Mixed MPE, $\bar{K} = 2$

Parameters	$u_Y \in \{0.25, 0.75\}$	$u_Y \in \{0.15, 0.85\}$
$\delta = 0.9$, $d = 1$		
$\delta = 0.9$, $d = 2$		
$\delta = 0.75$, $d = 1$		
$\delta = 0.75$, $d = 2$		

A2.2 Parameter Comparisons for the Dynamic Basin of Attraction of Grim Trigger against All-Y

Parameters	$\bar{K} = 2$	$\bar{K} = 3$
$\delta = 0.75$ $u_Y \in \{0.25, 0.75\}$		
$\delta = 0.75$ $u_Y \in \{0.15, 0.85\}$		
$\delta = 0.9$ $u_Y \in \{0.15, 0.85\}$		
$\delta = 0.9$ $u_Y \in \{0.25, 0.75\}$		

Appendix A3: History-dependent Strategies and Derivation of the Dynamic Basin

A3.1 Equivalent History-dependent Rules and Subgame Perfection

Markov Perfect Equilibria are state-based strategies that depend only on the current state, regardless of the history. By contrast, purely history-dependent strategies condition only on past play, while hybrid strategies condition on both the public state and the realized history. Consider the following four candidate rules. Two are purely history-dependent: (i) Grim Trigger independent of the state, denoted GT, which starts with X and switches permanently to Y after the first group failure, and (ii) Tit-for-Tat independent of the state, denoted TFT, which starts with X and in each later round mimics the previous group outcome, choosing X after all-X and Y otherwise. The other two are hybrid state-and-history-dependent rules: (iii) a state-dependent Grim Trigger, denoted $GT(k)$, that uses the same trigger rule but allows a state-based override only at $k = \bar{K}$, and (iv) a state-dependent Tit-for-Tat, denoted $TFT(k)$, that follows the same group-outcome rule with the same $k = \bar{K}$ override.

Under the public monitoring and reset rule in the present game, these four labels are distinct descriptions of the same induced strategy. After any group failure, the state next period is $k = 0$. Hence the override at $k = \bar{K}$ never binds off the cooperative path. Moreover, once TFT prescribes Y after the first group failure, all members choose Y, so the public outcome remains not all-X forever and TFT also stays at Y forever. Therefore, along the path of play and after any unilateral deviation from it, all four rules prescribe the same actions; they can differ only after histories in which every member simultaneously returns to X following a failure.

These four rules are also all Subgame Perfect Equilibria. After any history at which the rule prescribes Y for all members, choosing X cannot produce all-X and does not change continuation play, so Y is a strict best response. Along a cooperative public history at state k, let $m \equiv \max\{\bar{K} - k, 1\}$ (truncated at one because an all-X outcome at $k = \bar{K} - 1$ already reaches protection). If all other members follow any one of the four rules above, then deviating to Y immediately triggers the continuation value

$$V_k^{AY} = \frac{\delta}{1-\delta} u_Y - d.$$

By contrast, if agent i also follows the rule and all others continue with X, the cooperative continuation value is

$$W_m = \frac{\delta}{1-\delta} - d + (1+d)\delta^{m-1}, \qquad m \geq 1.$$

Therefore,

$$W_m - V_k^{AY} = \frac{\delta(1-u_Y)}{1-\delta} + (1+d)\delta^{m-1} > 0,$$

so no profitable one-shot deviation exists on the cooperative path either. Hence all four candidate rules are SPE. Because these four rules are behaviorally identical in this environment, I use GT as shorthand for this common induced strategy.

A3.2 Derivation of the Dynamic Basin

As in Section 2.3, agent i believes that each other member $j \neq i$ is a *Grim Trigger* type with probability p and an All-Y type with probability $1-p$. Agent i's continuation value as an All-Y type is

$$V_k^{AY} = V_0^{AY} = \delta\big(u_Y + V_0^{AY}\big) + (1-\delta)(-d).$$

Hence,

$$V_k^{AY} = \frac{\delta}{1-\delta} u_Y - d.$$

Agent i's continuation value as a *Grim Trigger* type has two components. First, if all other $n-1$ members are also *Grim Trigger*, which occurs with probability p^{n-1}, then all members coordinate on X until the threshold is reached. Denote $m \equiv \max\{\bar{K} - k, 1\}$ as the distance from the current state k to the threshold $\bar{K}$. The continuation value of the first part is therefore W_m, the same cooperative value as in Section A3.1. It satisfies

$$W_0 = W_1 = \frac{1}{1-\delta},$$

and for $2 \leq m \leq \bar{K}$,

$$W_m = \delta\big(1 + W_{m-1}\big) + (1-\delta)(-d).$$

Solving this recursion gives

$$W_m = \frac{\delta}{1-\delta} - d + (1+d)\delta^{m-1}.$$

Second, if at least one other member is an All-Y type, which occurs with probability $1-p^{n-1}$, then coordination on X fails immediately. In that case, agent i's current-stage payoff from choosing X is 0, and from the next period onward all Grim Trigger types switch to Y. Hence,

the continuation value of the second part is

$$\delta V_k^{AY} + (1-\delta)(-d) = \frac{\delta^2}{1-\delta} u_Y - d.$$

Therefore,

$$V_k^{GT} = p^{n-1} W_m + \left(1 - p^{n-1}\right) \left(\frac{\delta^2}{1-\delta} u_Y - d \right).$$

Solving $V_k^{GT} \geq V_k^{AY}$ gives

$$p^{n-1} \geq \frac{\delta u_Y}{\frac{\delta(1-\delta u_Y)}{1-\delta} + (1+d)\delta^{m-1}}.$$

Dividing the numerator and denominator by δ,

$$p \geq p_k^\star \equiv \left(\frac{u_Y}{\frac{1-\delta u_Y}{1-\delta} + (1+d)\delta^{m-2}} \right)^{\frac{1}{n-1}}.$$

A3.3 Equivalence with the Mixed-MPE Cutoff at $k = 0$

At $k = 0$, the distance to protection is $m = \max\{\bar{K}, 1\} = \bar{K}$. Multiplying the numerator and denominator of $(p_0^\star)^{n-1}$ by $(1-\delta)$, and for $\bar{K} = 1$ also by δ,

$$\begin{aligned}
\bar{K} = 1: \quad (p_0^\star)^{n-1} &= \frac{\delta(1-\delta)u_Y}{\delta(1-\delta u_Y) + (1+d)(1-\delta)} = \frac{\delta(1-\delta)u_Y}{1 - \delta^2 u_Y + (1-\delta)d}, \\
\bar{K} = 2: \quad (p_0^\star)^{n-1} &= \frac{(1-\delta)u_Y}{1 - \delta u_Y + (1+d)(1-\delta)} = \frac{(1-\delta)u_Y}{(2-\delta) - \delta u_Y + (1-\delta)d}, \\
\bar{K} = 3: \quad (p_0^\star)^{n-1} &= \frac{(1-\delta)u_Y}{1 - \delta u_Y + (1+d)\delta(1-\delta)} = \frac{(1-\delta)u_Y}{1 + \delta(1-\delta)(1+d) - \delta u_Y}.
\end{aligned}$$

These equal σ_0^{n-1} in the $(m1)$, $(m11)$, and $(m111)$ MPEs (Figure 1), so $p_0^\star = \sigma_0$ and $\mathcal{DB}_0 = 1 - \sigma_0$.

Appendix B1: Additional Empirical Results for p(X)

Table B1.1: Monotonicity in State: Alternative Specifications with Participant Controls

	State increments			Linear state		
	(1)	(2)	(3)	(4)	(5)	(6)
State = 1 minus State = 0	0.063***	0.063***	0.063***			
	(0.015)	(0.015)	(0.015)			
State = 2 minus State = 1	0.032*	0.032*	0.032*			
	(0.022)	(0.022)	(0.022)			
Difference between increments	0.031*	0.031*	0.031*			
	(0.018)	(0.018)	(0.018)			
State				0.049***	0.049***	0.049***
				(0.016)	(0.016)	(0.016)
Risk-averse		-0.021	-0.022		-0.021	-0.022
		(0.013)	(0.014)		(0.013)	(0.014)
Prosocial		0.037*	0.035		0.038*	0.036
		(0.018)	(0.021)		(0.019)	(0.021)
Gender (1=Male)			-0.008			-0.008
			(0.020)			(0.020)
Major (1=Engineering/Science)			0.019			0.020
			(0.020)			(0.019)
High school language (1=English)			0.022			0.022
			(0.022)			(0.021)
Group-by-match fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Round-within-match fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,812	7,812	7,812	7,812	7,812	7,812
R^2	0.567	0.569	0.570	0.566	0.569	0.570

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Session-clustered standard errors are in parentheses. Equal weight per session. Inference uses 5,000 wild-cluster-bootstrap-t replications.

As a secondary analysis, Table B1.1 reports alternative specifications that include elicited risk and social preferences and other participant characteristics. Columns (1)-(3) parameterize the state variable in terms of adjacent increments. The estimated increase in the probability assigned to X from $k = 0$ to $k = 1$ is 0.063, while the additional increase from $k = 1$ to $k = 2$ is 0.032. Thus, the point estimates are monotonic in the state and indicate a larger increase from $k = 0$ to $k = 1$ than from $k = 1$ to $k = 2$. The difference between the two increments is 0.031. These estimates are essentially unchanged as participant controls are added across columns (1)-(3). Columns (4)-(6) provide a complementary specification in which the state enters linearly. The estimated coefficient on k is 0.049, implying that a one-unit increase in the state is associated with an approximately 5 percentage-point increase in the probability assigned to X. The estimate is stable across specifications and statistically significant at the 1% level. Risk aversion is not significantly associated with $p(X)$, while

prosociality is positively associated with $p(X)$, although the latter relationship is only weakly statistically significant and is sensitive to the inclusion of additional participant controls.

Table B1.2: Initial versus Post-Reset Behavior at k=0

	Probability assigned to X				
	Pooled	2l	2h	4l	4h
	(1)	(2)	(3)	(4)	(5)
Post-reset $k = 0$	-0.052	0.089**	-0.015	-0.025	-0.226***
	(0.032)	(0.033)	(0.051)	(0.015)	(0.025)
Risk-averse	-0.042**	-0.008	-0.092	-0.042	0.001
	(0.021)	(0.061)	(0.057)	(0.039)	(0.049)
Prosocial	0.063*	0.128**	-0.021	0.043***	0.118
	(0.030)	(0.053)	(0.033)	(0.026)	(0.072)
Male	-0.008	0.065	-0.010	-0.038	-0.042
	(0.025)	(0.053)	(0.047)	(0.041)	(0.062)
Major	0.021	0.062	0.039	0.055**	-0.052
(1=Engineering/Science)	(0.026)	(0.052)	(0.062)	(0.025)	(0.053)
High school language	0.045	-0.014	0.021	0.166	-0.018
(1=English)	(0.025)	(0.064)	(0.029)	(0.047)	(0.046)
Group-by-match fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	4,000	554	714	1,000	1,732
Subjects	164	68	68	96	96
Sessions	12	6	6	6	6
Groups	580	170	170	120	120
R^2	0.518	0.345	0.622	0.338	0.426

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. SEs are clustered by session. All columns restrict the sample to the current state $k = 0$; the initial $k = 0$ is omitted. Treatment columns are estimated separately. Groups are distinct group-by-match cells.

Table B1.2 compares behavior at the initial $k = 0$ state with behavior at $k = 0$ following a reset. In the pooled specification, the post-reset coefficient is -0.052 and is not statistically significant, so I cannot reject equality between initial and post-reset behavior. The treatment-specific estimates, however, reveal heterogeneity: the post-reset effect is positive in treatment 2l (0.089, $p < 0.05$), close to zero in 2h and 4l, and strongly negative in 4h (-0.226, $p < 0.01$). Because each treatment-specific regression relies on six sessions, these treatment-level differences should be interpreted cautiously.

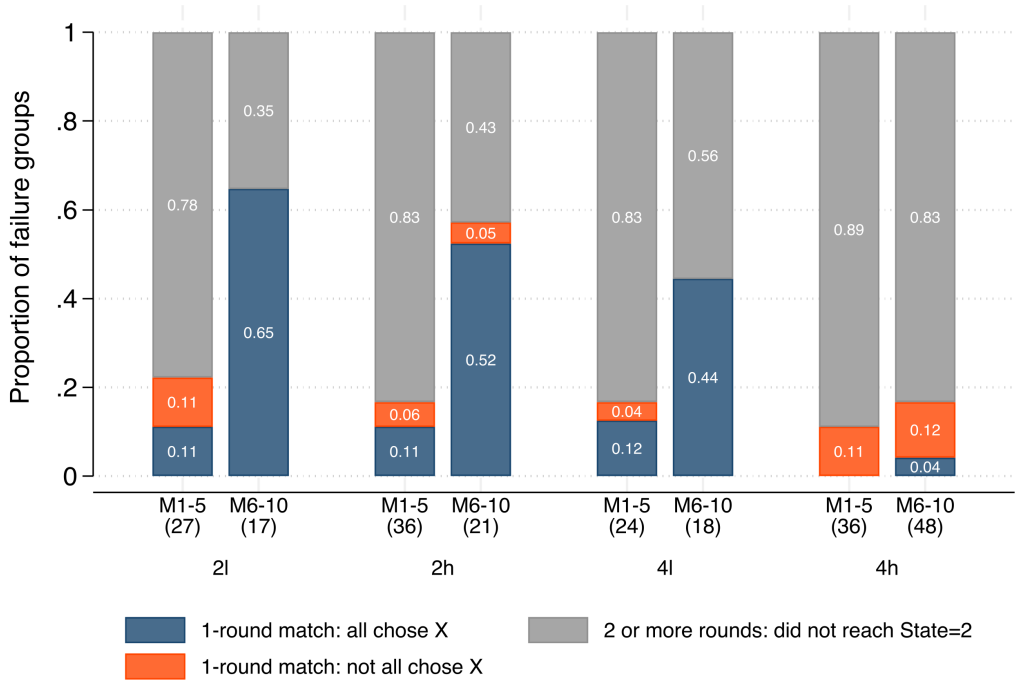


Figure B1.1: Composition of Groups That Do Not Reach the Threshold ($k = 2$)

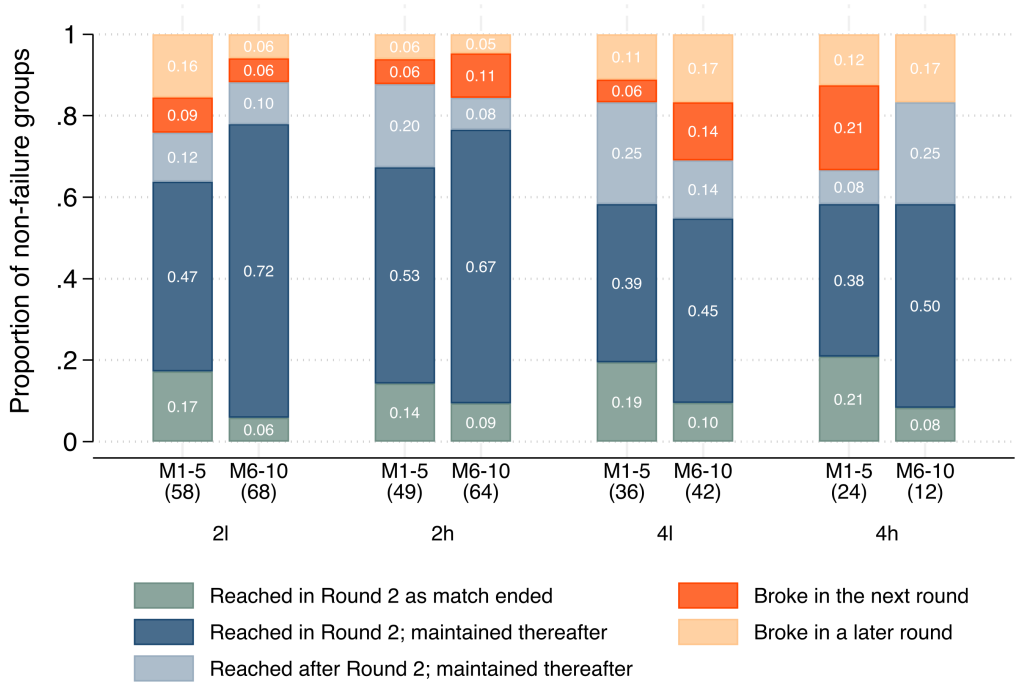


Figure B1.2: Outcomes among Groups That Reached the Threshold ($k = 2$)

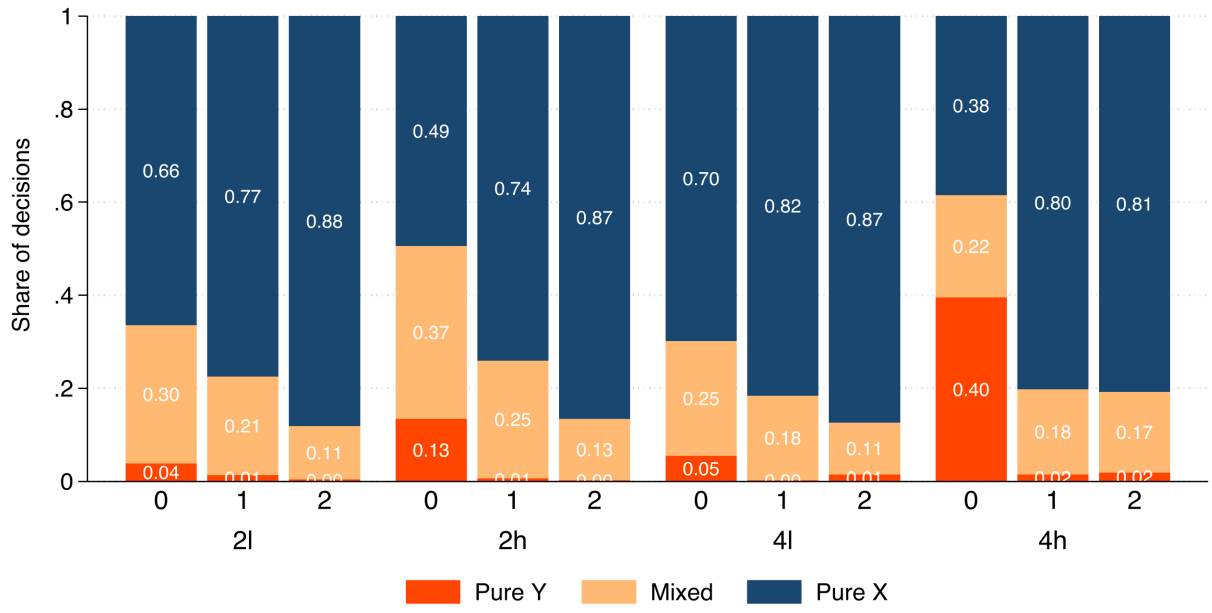


Figure B1.3: Decisions by State and Treatment

Table B1.3: Balance of Subject Characteristics: $n = 4$ Treatment and Baseline

Subject Characteristic	Treatment		Baseline		t-statistic
	Mean	(SE)	Mean	(SE)	
Gender (1=Male)	0.49	(0.03)	0.63	(0.06)	-1.96
Major (1=Engineering/Science)	0.57	(0.06)	0.41	(0.16)	0.99
High school language (1=English)	0.88	(0.05)	0.97	(0.03)	-1.63
Risk averse (1=Yes)	0.44	(0.04)	0.41	(0.09)	0.31
Social Preference (1=Prosocial)	0.61	(0.03)	0.53	(0.03)	1.81
Sessions	6		2		
Subjects	96		32		

Notes: Each characteristic is first averaged within an independent session. Standard errors are standard errors of the session means. The final column reports the unequal-variance t-statistic for treatment minus baseline as a descriptive balance diagnostic.

Appendix B2: Additional Empirical Results for All-X

This appendix reports corresponding analyses using realized all-X coordination as a secondary group-level outcome.

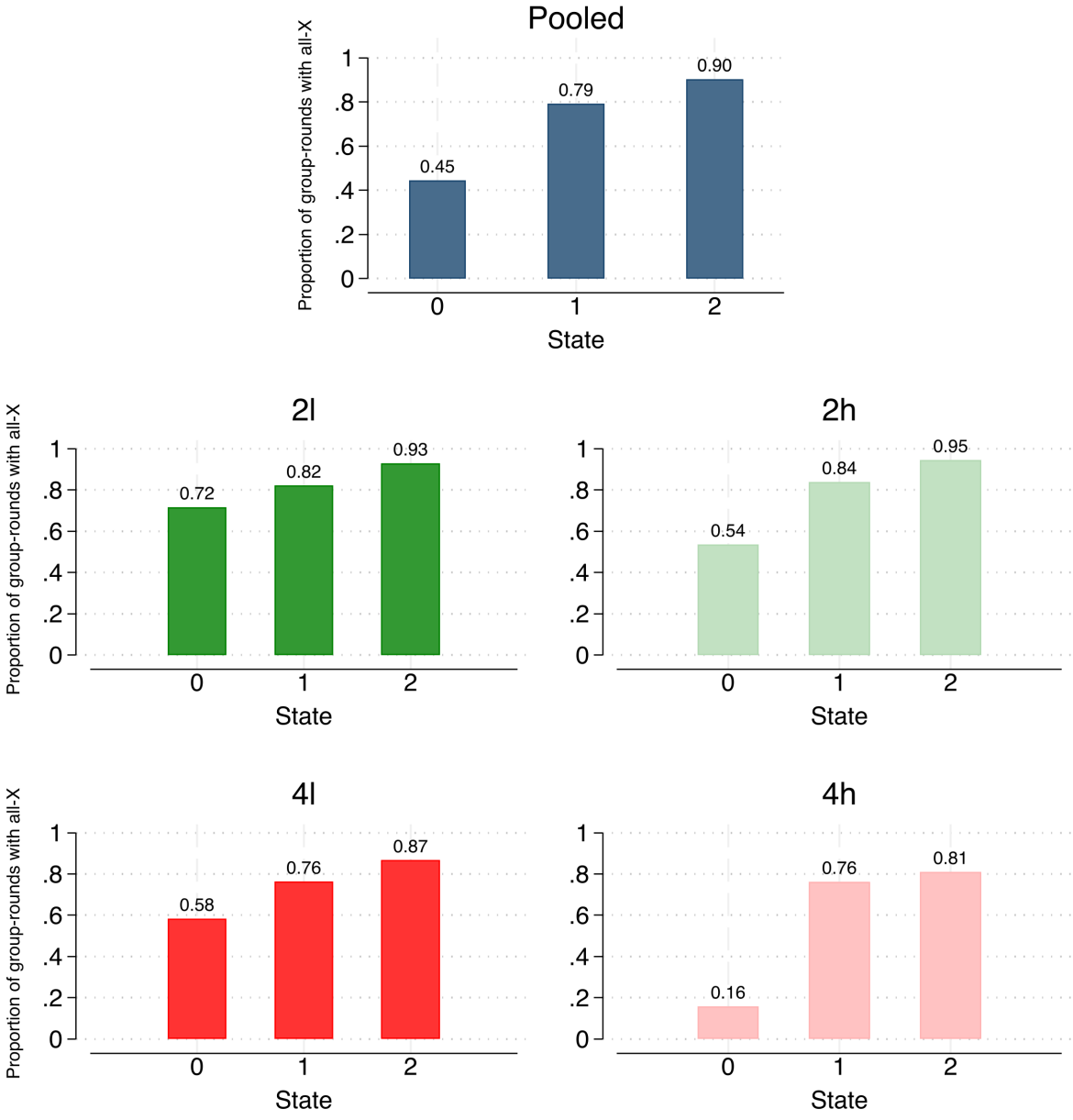
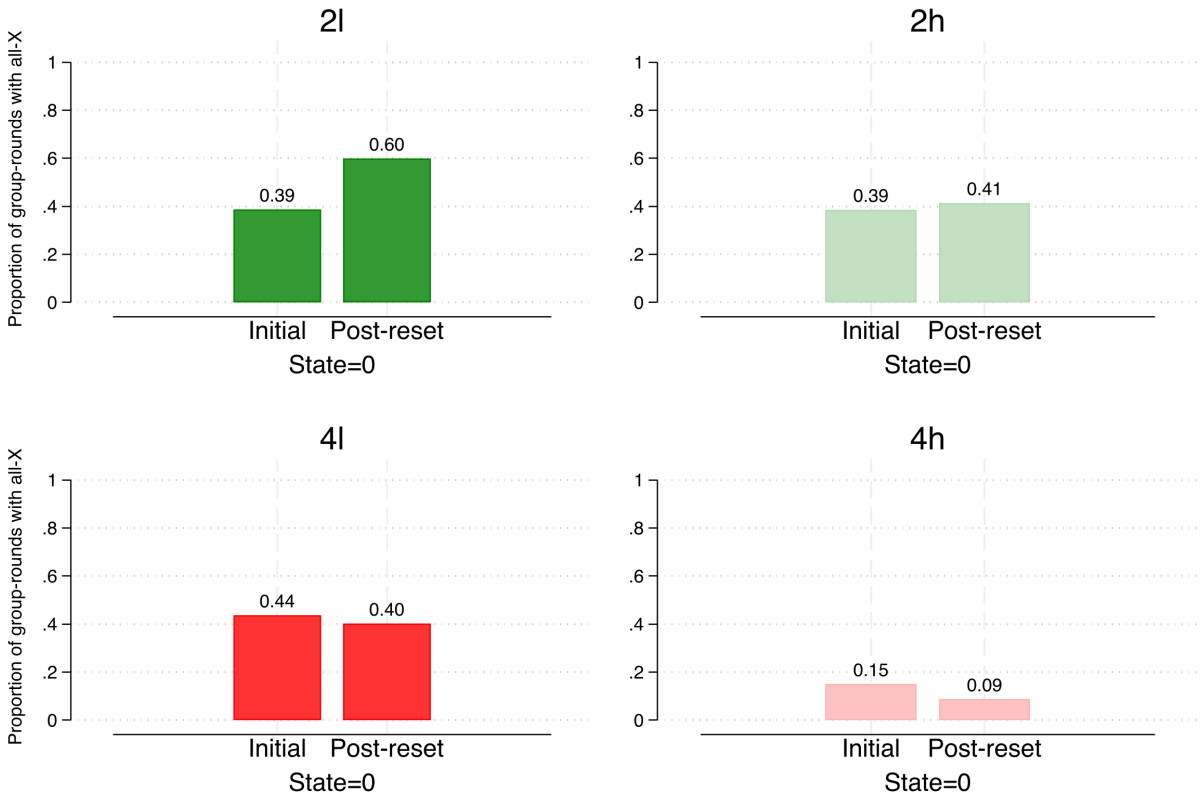


Figure B2.1: Proportion of Group-Rounds with All-X Coordination by State



Notes: The sample is restricted to group-by-match cells containing both the initial Round-1 $k = 0$ observation and at least one post-reset $k = 0$ observation.

Figure B2.2: Proportion of Group-Rounds with All-X Coordination at Initial and Post-Reset $k = 0$ by Treatment

Table B2.1: Group Size and Realized All-X Coordination

	Low u_Y		High u_Y	
	Unadjusted (1)	Adjusted (2)	Unadjusted (3)	Adjusted (4)
$n = 2$ minus $n = 4$	0.109	0.109	0.415**	0.415**
	(0.082)	(0.060)	(0.112)	(0.101)
High-first	No	Yes	No	Yes
Sequence fixed effects	No	Yes	No	Yes
Sessions	12	12	12	12

Notes: The binary outcome equals one when all group members choose X. One observation is retained per group-round, then averaged within each session and payoff condition. Standard errors are in parentheses. Adjusted columns include High-first and sequence fixed effects. Stars use one-sided exact randomization tests (64 assignments). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Positive estimates are in the direction predicted by Hypothesis 3a.

Table B2.2: Safe-Option Payoff and Realized All-X Coordination

	$n = 2$			$n = 4$		
	(1)	(2)	(3)	(4)	(5)	(6)
Low u_Y minus High u_Y	0.112**	0.112	0.169	0.417***	0.417***	0.293
	(0.040)	(0.091)	(0.041)	(0.071)	(0.073)	(0.180)
Session fixed effects	Yes	Yes		Yes	Yes	
Absolute match-number FE	Yes		Yes	Yes		Yes
Relative match-number FE		Yes			Yes	
First-half sample	No	No	Yes	No	No	Yes
Group-rounds	1,618	1,618	790	1,144	1,144	560
Sessions	6	6	6	6	6	6
R^2	0.103	0.048	0.044	0.286	0.283	0.090

Notes: The binary outcome equals one when all group members choose X; one observation is retained per group-round. Full-sample columns use equal-session analytic weights, session fixed effects, and session-clustered standard errors. Stars use one-sided wild-cluster-bootstrap-t tests in columns (1), (2), (4), and (5), and one-sided exact session-level permutation tests in columns (3) and (6). First-half columns use equal-session weights and behavior before exposure to the alternative payoff. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Appendix C1: Experimental Instructions and Interfaces

The decisions, random draws, player tags, outcomes, and payment amounts shown in the screenshots are illustrative examples generated for interface documentation.

C1.1 Experiment Overview

Your payment for this experiment consists of earnings from the comprehension quiz and the money you accumulate during this experiment. The amount of money you accumulate will depend partly on your actions and partly on the actions of other participants. This money will be paid at the end of the experiment, privately and in cash.

It is important that during the experiment you **remain silent**. If you have a question or need assistance of any kind, **please raise your hand, but do not speak** — and an experiment administrator will come to you, and you may then whisper your question. In addition, **please turn off your cell phones and put them away now**. Anybody who breaks these rules will be asked to leave.

All participants in this room are given exactly the same instructions, and we will not deceive you in any way.

Agenda

- Instructions for Part 1
- Comprehension Quiz
- Part 1
- Part 2 (with Instructions)
- Part 3 (with Instructions)
- Survey

Next

C1.2 Instructions for Part 1 ($n = 2$ treatment sessions)

Please read the instructions carefully. A comprehension quiz follows. **Each correct answer earns you $0.50.**

[How a Match Works]

Part 1 consists of 10 matches. At the beginning of each match, you will be randomly grouped with 1 other participant in this room. You will then play a number of rounds with that 1 participant. This is what we call a "match". The two of you form a "group". Once a match ends, you will be randomly grouped with 1 other participant for a new match. You will not be able to identify who you've interacted with in previous or future matches.

The length of a match, that is, the number of rounds in a match, is randomly determined as follows: **After each round, there is an 80% probability that the match will continue for at least another round.** Specifically, after each round, whether the match continues for another round will be determined by a spinner with 80% white area and 20% grey area. If the arrow stops in the white area, the match will continue for at least another round; Otherwise, it will end. For example, if you are in round 2, the probability that there will be a third round is 80%, and if you are in round 5, the probability that there will be a sixth round is also 80%. At any point in a match, the probability that the match will continue is 80%.



✓ The match continues to the next round.

(arrow in white area)



✕ The match ends at the end of this round.

(arrow in grey area)

[Payment for Part 1]

All payoffs in Part 1 are in units of experimental points and will be converted to US Dollars at the rate of 400 Experimental Points = 1 US Dollar. You will have an endowment of 100 points at the beginning of each match. The total earnings of a match are the sum of the endowment and the payoffs received in each round of that match. **Your payment for Part 1 is the sum of your total earnings across all 10 matches.**

[Actions and Payoffs]

Each round, you and 1 other matched participant will each choose X or Y individually and simultaneously. Your round payoff is determined by your action and the group outcome, as shown in the payoff table. The group outcome shows each member's action in the order of Players 1 and 2 (P1 P2). The group player tag is randomly assigned to you at the beginning of each match and remains the same within that match.

Note: The payoffs here are for illustration only. You will be informed of the real payoffs at the beginning of each match.

Your action	Group outcome (P1 P2)	Your payoff
X	X X	100
	At least one action is Y	0
Y	At least one action is Y	50

Rather than directly choosing between X and Y, you assign probabilities to the two actions, and the computer decides which action to take based on these probabilities. You can think of this as allocating an area between the two actions. Below are 100 grids for you to allocate between Yellow (X) and Blue (Y) using a slider. Drag the slider to choose how many grids to allocate to each action. After you confirm your allocation, the computer will randomly draw an integer number between 1 and 100.

- If the number is in the Yellow area, your action is X.
- If the number is in the Blue area, your action is Y.

Let's try it. **Click anywhere on the slider to activate it.** Then, drag the slider thumb to your preferred position and click the Confirm Decision button to see the result. This is a practice, so you may drag the slider and try as many times as you like. **In real matches, however, you CANNOT change the probabilities once you click the Confirm Decision button.** Click Next when you are ready to move on.



[The Public Counter]

The public counter tracks the number of consecutive rounds in which the group outcome is X X among the two most recent rounds. You will see the public counter at the beginning of each round before you drag the slider. The public counter is initially set to 0 and shown at the beginning of Round 1, since there is no previous round. At the end of each round, the public counter is updated based on the group outcome and carried over to the beginning of the next round:

- If group outcome is X X, the counter **increases by 1, up to a maximum of 2**.
- If at least one action in the group outcome is Y, the counter **resets to 0**.



[Payoff in the Last Round of a Match]

As stated before, your round payoffs are determined by the payoff table. **However, your payoff in the last round of a match may be affected by the public counter.** At the end of the last round:

- If the counter **equals 2**, your payoff in the last round is determined by the payoff table.
- If the counter **is below 2**, everyone's payoff in the last round **is overridden to -100**, regardless of what is shown in the payoff table.

Note that you don't know whether a round is the last round of that match when you assign probabilities to X and Y in that round.

C1.3 Instructions for Part 1 ($n = 4$ treatment sessions)

Please read the instructions carefully. A comprehension quiz follows. **Each correct answer earns you $0.50.**

[How a Match Works]

Part 1 consists of 10 matches. At the beginning of each match, you will be randomly grouped with 3 other participants in this room. You will then play a number of rounds with those 3 participants. This is what we call a "match". The four of you form a "group". Once a match ends, you will be randomly grouped with 3 other participants for a new match. You will not be able to identify who you've interacted with in previous or future matches.

The length of a match, that is, the number of rounds in a match, is randomly determined as follows: **After each round, there is an 80% probability that the match will continue for at least another round.** Specifically, after each round, whether the match continues for another round will be determined by a spinner with 80% white area and 20% grey area. If the arrow stops in the white area, the match will continue for at least another round; Otherwise, it will end. For example, if you are in round 2, the probability that there will be a third round is 80%, and if you are in round 5, the probability that there will be a sixth round is also 80%. At any point in a match, the probability that the match will continue is 80%.



✓ The match continues to the next round.

(arrow in white area)



✕ The match ends at the end of this round.

(arrow in grey area)

[Payment for Part 1]

All payoffs in Part 1 are in units of experimental points and will be converted to US Dollars at the rate of 400 Experimental Points = 1 US Dollar. You will have an endowment of 100 points at the beginning of each match. The total earnings of a match are the sum of the endowment and the payoffs received in each round of that match. **Your payment for Part 1 is the sum of your total earnings across all 10 matches.**

[Actions and Payoffs]

Each round, you and 3 other matched participants will each choose X or Y individually and simultaneously. Your round payoff is determined by your action and the group outcome, as shown in the payoff table. The group outcome shows each member's action in the order of Players 1, 2, 3, and 4 (P1 P2 P3 P4). The group player tag is randomly assigned to you at the beginning of each match and remains the same within that match.

Note: The payoffs here are for illustration only. You will be informed of the real payoffs at the beginning of each match.

Your action	Group outcome (P1 P2 P3 P4)	Your payoff
X	X X X X	100
	At least one action is Y	0
Y	At least one action is Y	50

Rather than directly choosing between X and Y, you assign probabilities to the two actions, and the computer decides which action to take based on these probabilities. You can think of this as allocating an area between the two actions. Below are 100 grids for you to allocate between Yellow (X) and Blue (Y) using a slider. Drag the slider to choose how many grids to allocate to each action. After you confirm your allocation, the computer will randomly draw an integer number between 1 and 100.

- If the number is in the Yellow area, your action is X.
- If the number is in the Blue area, your action is Y.

Let's try it. **Click anywhere on the slider to activate it.** Then, drag the slider thumb to your preferred position and click the Confirm Decision button to see the result. This is a practice, so you may drag the slider and try as many times as you like. **In real matches, however, you CANNOT change the probabilities once you click the Confirm Decision button.** Click Next when you are ready to move on.



[The Public Counter]

The public counter tracks the number of consecutive rounds in which the group outcome is X X X X among the two most recent rounds. You will see the public counter at the beginning of each round before you drag the slider. The public counter is initially set to 0 and shown at the beginning of Round 1, since there is no previous round. At the end of each round, the public counter is updated based on the group outcome and carried over to the beginning of the next round:

- If group outcome is X X X X, the counter **increases by 1, up to a maximum of 2**.
- If at least one action in the group outcome is Y, the counter **resets to 0**.



[Payoff in the Last Round of a Match]

As stated before, your round payoffs are determined by the payoff table. **However, your payoff in the last round of a match may be affected by the public counter.** At the end of the last round:

- If the counter **equals 2**, your payoff in the last round is determined by the payoff table.
- If the counter **is below 2**, everyone's payoff in the last round **is overridden to -100**, regardless of what is shown in the payoff table.

Note that you don't know whether a round is the last round of that match when you assign probabilities to X and Y in that round.

C1.4 Instructions for Part 1 ($n = 2$ baseline sessions)

Please read the instructions carefully. A comprehension quiz follows. **Each correct answer earns you $0.50.**

[How a Match Works]

Part 1 consists of 10 matches. At the beginning of each match, you will be randomly grouped with 1 other participant in this room. You will then play a number of rounds with that 1 participant. This is what we call a "match". The two of you form a "group". Once a match ends, you will be randomly grouped with 1 other participant for a new match. You will not be able to identify who you've interacted with in previous or future matches.

The length of a match, that is, the number of rounds in a match, is randomly determined as follows: **After each round, there is an 80% probability that the match will continue for at least another round.** Specifically, after each round, whether the match continues for another round will be determined by a spinner with 80% white area and 20% grey area. If the arrow stops in the white area, the match will continue for at least another round; Otherwise, it will end. For example, if you are in round 2, the probability that there will be a third round is 80%, and if you are in round 5, the probability that there will be a sixth round is also 80%. At any point in a match, the probability that the match will continue is 80%.



✓ The match continues to the next round.

(arrow in white area)



✕ The match ends at the end of this round.

(arrow in grey area)

[Payment for Part 1]

All payoffs in Part 1 are in units of experimental points and will be converted to US Dollars at the rate of 400 Experimental Points = 1 US Dollar. You will have an endowment of 100 points at the beginning of each match. The total earnings of a match are the sum of the endowment and the payoffs received in each round of that match. **Your payment for Part 1 is the sum of your total earnings across all 10 matches.**

[Actions and Payoffs]

Each round, you and 1 other matched participant will each choose X or Y individually and simultaneously. Your round payoff is determined by your action and the group outcome, as shown in the payoff table. The group outcome shows each member's action in the order of Players 1 and 2 (P1 P2). The group player tag is randomly assigned to you at the beginning of each match and remains the same within that match.

Note: The payoffs here are for illustration only. You will be informed of the real payoffs at the beginning of each match.

Your action	Group outcome (P1 P2)	Your payoff
X	X X	100
	At least one action is Y	0
Y	At least one action is Y	50

Rather than directly choosing between X and Y, you assign probabilities to the two actions, and the computer decides which action to take based on these probabilities. You can think of this as allocating an area between the two actions. Below are 100 grids for you to allocate between Yellow (X) and Blue (Y) using a slider. Drag the slider to choose how many grids to allocate to each action. After you confirm your allocation, the computer will randomly draw an integer number between 1 and 100.

- If the number is in the Yellow area, your action is X.
- If the number is in the Blue area, your action is Y.

Let's try it. **Click anywhere on the slider to activate it.** Then, drag the slider thumb to your preferred position and click the Confirm Decision button to see the result. This is a practice, so you may drag the slider and try as many times as you like. **In real matches, however, you CANNOT change the probabilities once you click the Confirm Decision button.** Click Next when you are ready to move on.


X (Yellow)
Y (Blue)
1 2 3 4 5 6 7 8 9 10
11 12 13 14 15 16 17 18 19 20
21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40
41 42 43 44 45 46 47 48 49 50
51 52 53 54 55 56 57 58 59 60
61 62 63 64 65 66 67 68 69 70
71 72 73 74 75 76 77 78 79 80
81 82 83 84 85 86 87 88 89 90
91 92 93 94 95 96 97 98 99 100
Yellow (X) –%
Blue (Y) –%
100%
Y
100%
X
If you want to have action X for sure, drag the slider to the right end.
If you want to have action Y for sure, drag the slider to the left end.
The payoff listed in this practice is for illustration only.

C1.5 Instructions for Part 1 ($n = 4$ baseline sessions)

Please read the instructions carefully. A comprehension quiz follows. **Each correct answer earns you $0.50.**

[How a Match Works]

Part 1 consists of 10 matches. At the beginning of each match, you will be randomly grouped with 3 other participants in this room. You will then play a number of rounds with those 3 participants. This is what we call a "match". The four of you form a "group". Once a match ends, you will be randomly grouped with 3 other participants for a new match. You will not be able to identify who you've interacted with in previous or future matches.

The length of a match, that is, the number of rounds in a match, is randomly determined as follows: **After each round, there is an 80% probability that the match will continue for at least another round.** Specifically, after each round, whether the match continues for another round will be determined by a spinner with 80% white area and 20% grey area. If the arrow stops in the white area, the match will continue for at least another round; Otherwise, it will end. For example, if you are in round 2, the probability that there will be a third round is 80%, and if you are in round 5, the probability that there will be a sixth round is also 80%. At any point in a match, the probability that the match will continue is 80%.



✓ The match continues to the next round.

(arrow in white area)



✕ The match ends at the end of this round.

(arrow in grey area)

[Payment for Part 1]

All payoffs in Part 1 are in units of experimental points and will be converted to US Dollars at the rate of 400 Experimental Points = 1 US Dollar. You will have an endowment of 100 points at the beginning of each match. The total earnings of a match are the sum of the endowment and the payoffs received in each round of that match. **Your payment for Part 1 is the sum of your total earnings across all 10 matches.**

[Actions and Payoffs]

Each round, you and 3 other matched participants will each choose X or Y individually and simultaneously. Your round payoff is determined by your action and the group outcome, as shown in the payoff table. The group outcome shows each member's action in the order of Players 1, 2, 3, and 4 (P1 P2 P3 P4). The group player tag is randomly assigned to you at the beginning of each match and remains the same within that match.

Note: The payoffs here are for illustration only. You will be informed of the real payoffs at the beginning of each match.

Your action	Group outcome (P1 P2 P3 P4)	Your payoff
X	X X X X	100
	At least one action is Y	0
Y	At least one action is Y	50

Rather than directly choosing between X and Y, you assign probabilities to the two actions, and the computer decides which action to take based on these probabilities. You can think of this as allocating an area between the two actions. Below are 100 grids for you to allocate between Yellow (X) and Blue (Y) using a slider. Drag the slider to choose how many grids to allocate to each action. After you confirm your allocation, the computer will randomly draw an integer number between 1 and 100.

- If the number is in the Yellow area, your action is X.
- If the number is in the Blue area, your action is Y.

Let's try it. **Click anywhere on the slider to activate it.** Then, drag the slider thumb to your preferred position and click the Confirm Decision button to see the result. This is a practice, so you may drag the slider and try as many times as you like. **In real matches, however, you CANNOT change the probabilities once you click the Confirm Decision button.** Click Next when you are ready to move on.


X (Yellow)
Y (Blue)
1 2 3 4 5 6 7 8 9 10
11 12 13 14 15 16 17 18 19 20
21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40
41 42 43 44 45 46 47 48 49 50
51 52 53 54 55 56 57 58 59 60
61 62 63 64 65 66 67 68 69 70
71 72 73 74 75 76 77 78 79 80
81 82 83 84 85 86 87 88 89 90
91 92 93 94 95 96 97 98 99 100
Yellow (X) –%
Blue (Y) –%
100%
Y
100%
X
If you want to have action X for sure, drag the slider to the right end.
If you want to have action Y for sure, drag the slider to the left end.
The payoff listed in this practice is for illustration only.

C1.6 Comprehension Quiz

To avoid repetition, all screenshots in this part show interfaces where participants select incorrect answers; correct answers are marked in green. Some quiz questions apply only to particular sessions, as indicated by the figure titles. For example, "$n = 2$ sessions" refers to both $n = 2$ treatment and baseline sessions, while "baseline sessions" refers to both $n = 2$ and $n = 4$ baseline sessions. Figures without such labels apply to all sessions.

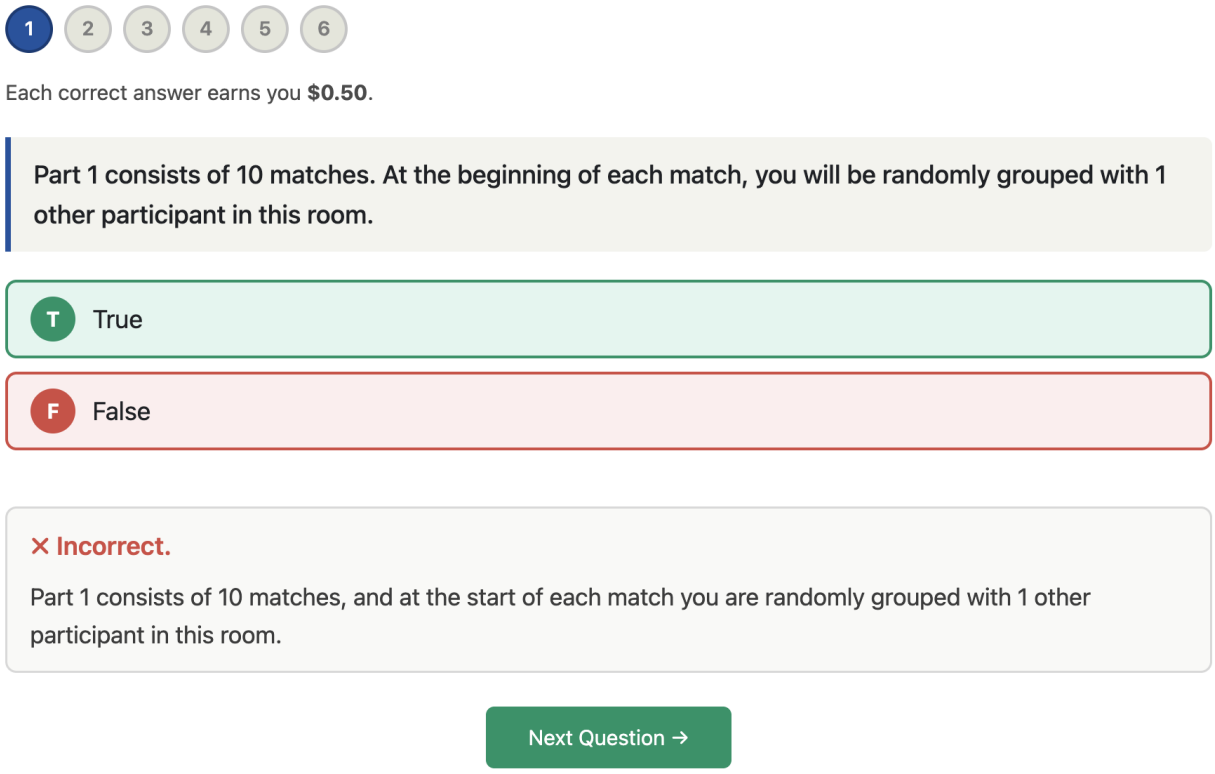


Figure C1.6.1: Question 1 ($n = 2$ sessions)

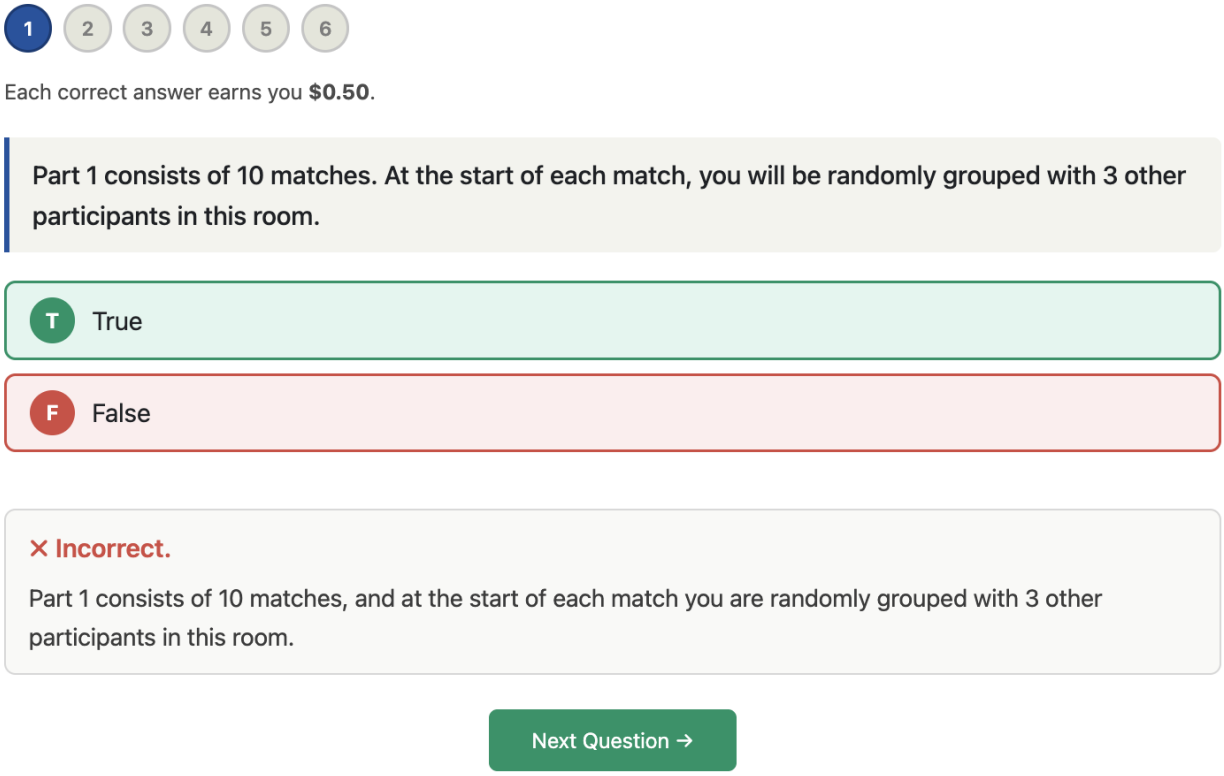


Figure C1.6.2: Question 1 ($n = 4$ sessions)

1 2 3 4 5 6

Each correct answer earns you **$0.50**.

At the end of each round, the computer spins an arrow on a 20% grey-80% white pie. Suppose in this round, the arrow stops in the white area. Then, the match ends at the end of this round.



T True

F False

✕ Incorrect.

If the arrow stops in the white area, the match will continue for at least another round; Otherwise, it will end. In this round the arrow stops in the white area, so the match does NOT end at the end of this round.

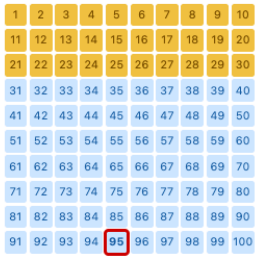
Next Question →

Figure C1.6.3: Question 2

1 2 3 4 5 6

Each correct answer earns you **$0.50**.

Suppose this is NOT the last round of a match. You allocate the 100 grids as 1-30 for Yellow (X) and 31-100 for Blue (Y). The integer drawn by the computer is 95. Which of the following options is correct?



Integer drawn: **95**

Your action	Group outcome (P1 P2)	Your payoff
X	X X	**100**
	At least one action is Y	**0**
Y	At least one action is Y	**50**

A Your action is Y, and your round payoff can be 100, 0, or 50.

B Your action is Y, and your round payoff is 50.

C Your action is X, and your round payoff is 100.

D Your action is X, and your round payoff is 0.

✕ Incorrect.

The integer drawn by the computer is 95, which is in the Blue area (grids 31-100), so your action is Y. At least one action in the group outcome is Y, so your round payoff is 50.

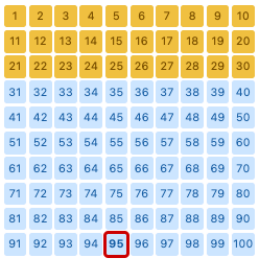
Next Question →

Figure C1.6.4: Question 3 ($n = 2$ sessions)

1 2 3 4 5 6

Each correct answer earns you **$0.50**.

Suppose this is NOT the last round of a match. You allocate the 100 grids as 1-30 for Yellow (X) and 31-100 for Blue (Y). The integer drawn by the computer is 95. Which of the following options is correct?



Integer drawn: **95**

Your action	Group outcome (P1 P2 P3 P4)	Your payoff
X	X X X X	100
	At least one action is Y	0
Y	At least one action is Y	50

A Your action is Y, and your round payoff can be 100, 0, or 50.

B Your action is Y, and your round payoff is 50.

C Your action is X, and your round payoff is 100.

D Your action is X, and your round payoff is 0.

✕ **Incorrect.**

The integer drawn by the computer is 95, which is in the Blue area (grids 31-100), so your action is Y. At least one action in the group outcome is Y, so your round payoff is 50.

Next Question →

Figure C1.6.5: Question 3 ($n = 4$ sessions)

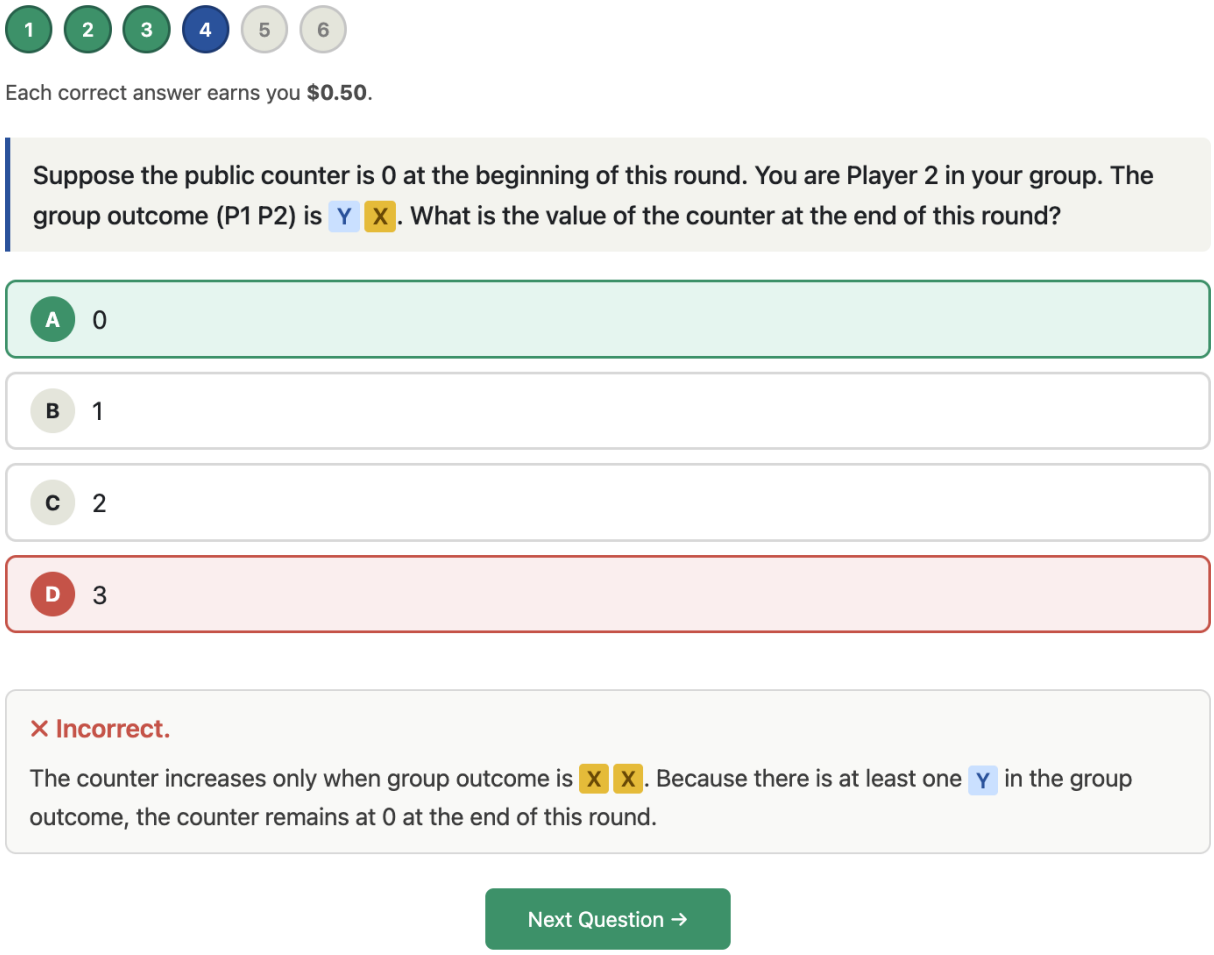


Figure C1.6.6: Question 4 ($n = 2$ treatment sessions)

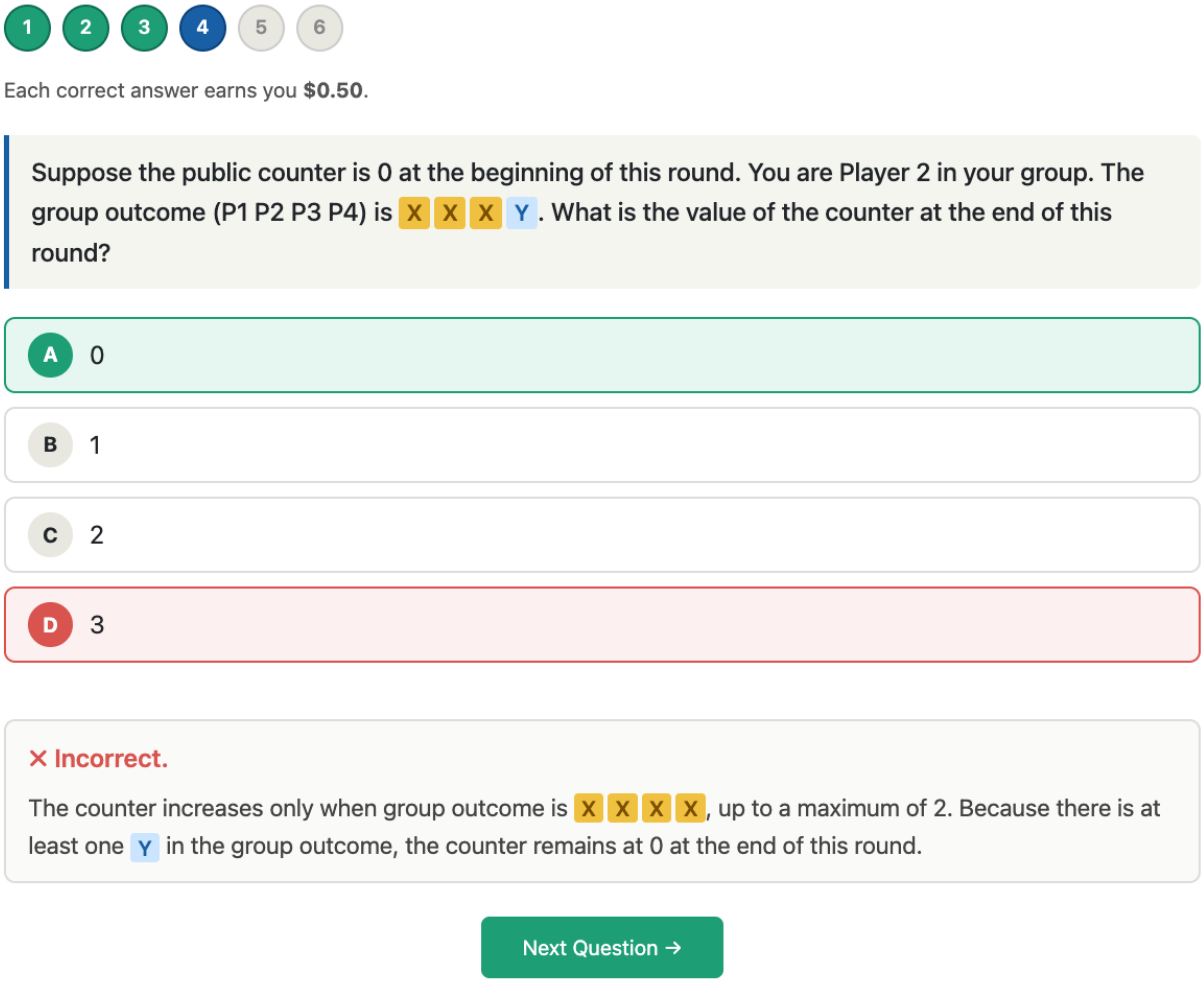


Figure C1.6.7: Question 4 ($n = 4$ treatment sessions)

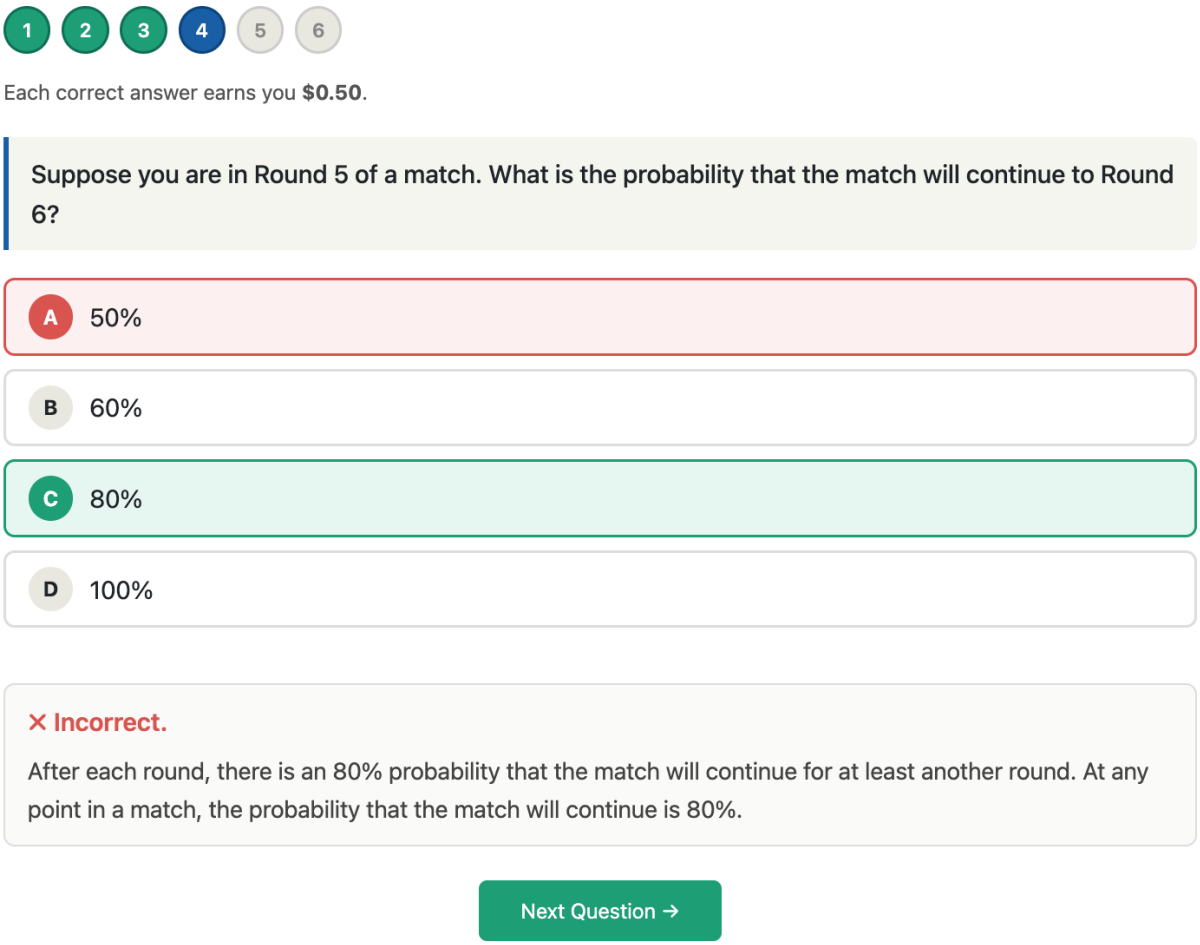


Figure C1.6.8: Question 4 (baseline sessions)

1 2 3 4 5 6

Each correct answer earns you **$0.50**.

Your action	Group outcome (P1 P2)	Your payoff
X	X X	100
	At least one action is Y	0
Y	At least one action is Y	50



Ends

Suppose the public counter is 0 at the beginning of this round, and the group outcome (P1 P2) is X X. You later find out that the spinner for this round stops in the grey area. What's your payoff in this round?

A 100

B 50

C 0

D −100

✕ Incorrect.

The arrow stops in the grey area, so this round turns out to be **the last round of this match**. The group outcome is X X, so the counter increases from 0 to 1 at the end of this round. The counter is below 2 at the end of the last round, so everyone's payoff of the last round **is overridden to −100**, regardless of what is shown in the payoff table.

Next Question →

Figure C1.6.9: Question 5 ($n = 2$ treatment sessions)

1 2 3 4 5 6

Each correct answer earns you **$0.50**.

Your action	Group outcome (P1 P2 P3 P4)	Your payoff
X	X X X X	100
	At least one action is Y	0
Y	At least one action is Y	50



Ends

Suppose the public counter is 0 at the beginning of this round, and the group outcome (P1 P2 P3 P4) is X X X X. You later find out that the spinner for this round stops in the grey area. What's your payoff in this round?

A 100

B 50

C 0

D −100

✕ **Incorrect.**

The arrow stops in the grey area, so this round turns out to be **the last round of this match**. The group outcome is X X X X, so the counter increases from 0 to 1 at the end of this round. The counter is below 2 at the end of the last round, so everyone's payoff of the last round **is overridden to −100**, regardless of what is shown in the payoff table.

Next Question →

Figure C1.6.10: Question 5 ($n = 4$ treatment sessions)

1 2 3 4 5 6

Each correct answer earns you **$0.50**.

Your action	Group outcome (P1 P2)	Your payoff
X	X X	100
	At least one action is Y	0
Y	At least one action is Y	50



Ends

Suppose you are Player 1 (P1) and in this round, the group outcome (P1 P2) is X X. You later find out that in this round, the arrow stops in the grey area. What's your payoff in this round?

A 100

B 50

C 0

D We don't know for sure.

✕ Incorrect.

Your round payoff is determined by your action and the group outcome, as shown in the payoff table. Your action is X, and the group outcome is X X, so your round payoff is 100. The arrow stops in the grey area, so the match ends at the end of this round.

Next Question →

Figure C1.6.11: Question 5 ($n = 2$ baseline sessions)

1 2 3 4 5 6

Each correct answer earns you **$0.50**.

Your action	Group outcome (P1 P2 P3 P4)	Your payoff
X	X X X X	100
	At least one action is Y	0
Y	At least one action is Y	50



Ends

Suppose you are Player 1 (P1) and in this round, the group outcome (P1 P2 P3 P4) is X X X X. You later find out that in this round, the arrow stops in the grey area. What's your payoff in this round?

A 100

B 50

C 0

D We don't know for sure.

✕ Incorrect.

Your round payoff is determined by your action and the group outcome, as shown in the payoff table. Your action is X, and the group outcome is X X X X, so your round payoff is 100. The arrow stops in the grey area, so the match ends at the end of this round.

Next Question →

Figure C1.6.12: Question 5 ($n = 4$ baseline sessions)

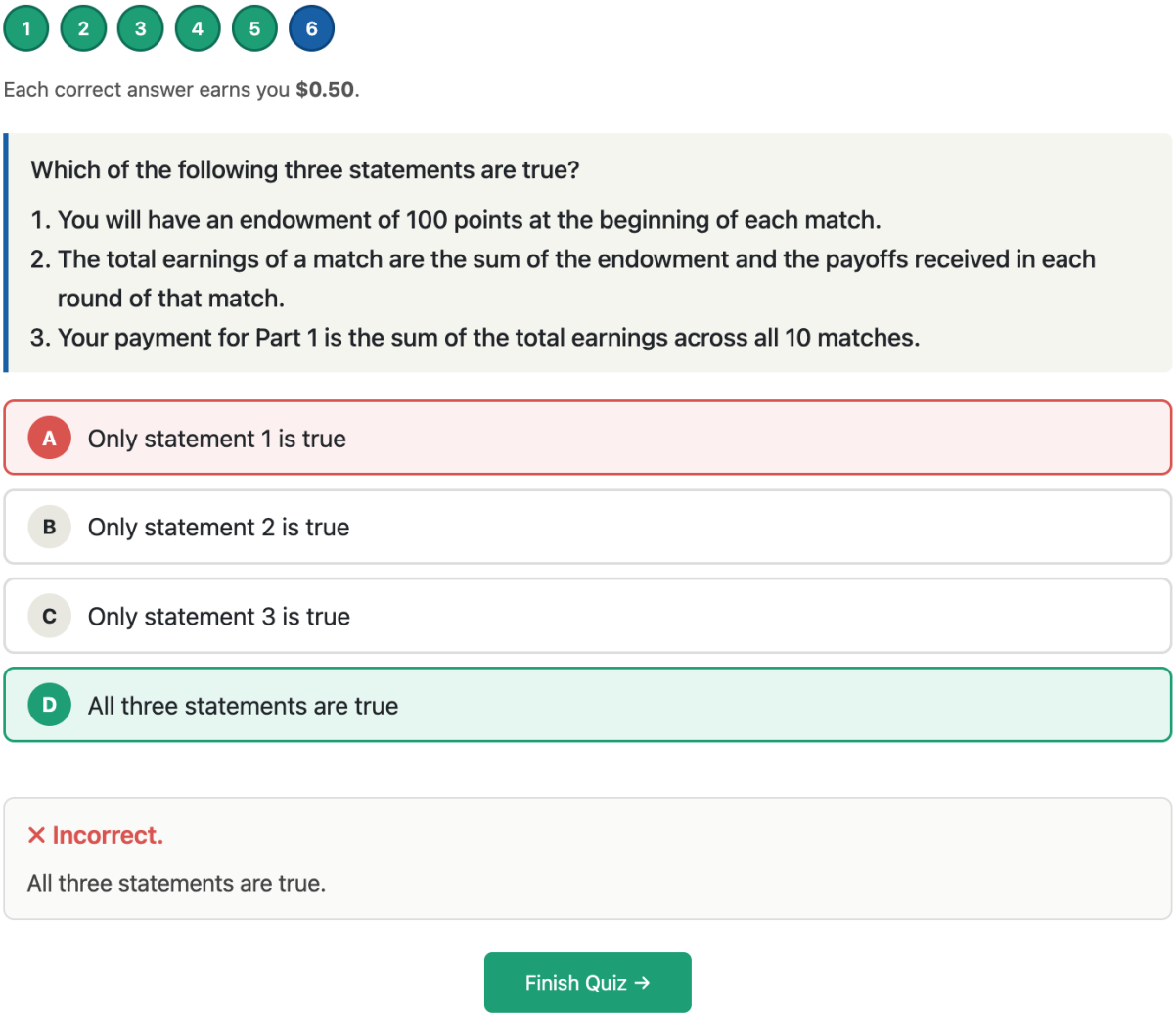


Figure C1.6.13: Question 6

C1.7 Part 1: 10 Matches ($n = 2$ sessions)

To reduce repetition, all screenshots in this part show the Part 1 interfaces in page order, for the $n = 2$ sessions only. Explanations are provided in figure titles or between figures.

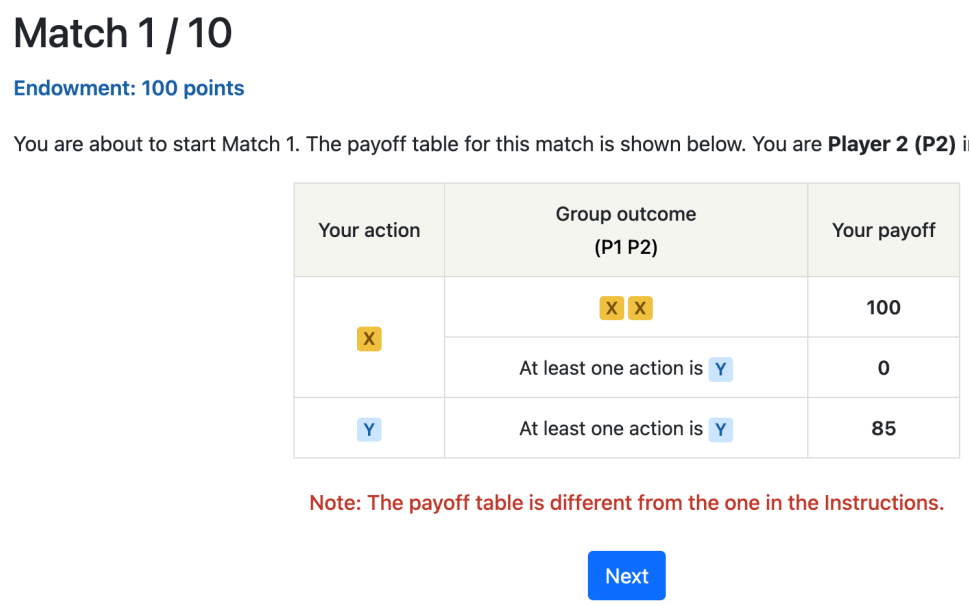


Figure C1.7.1: Welcome page for Matches 1-5 ($n = 2$ sessions)

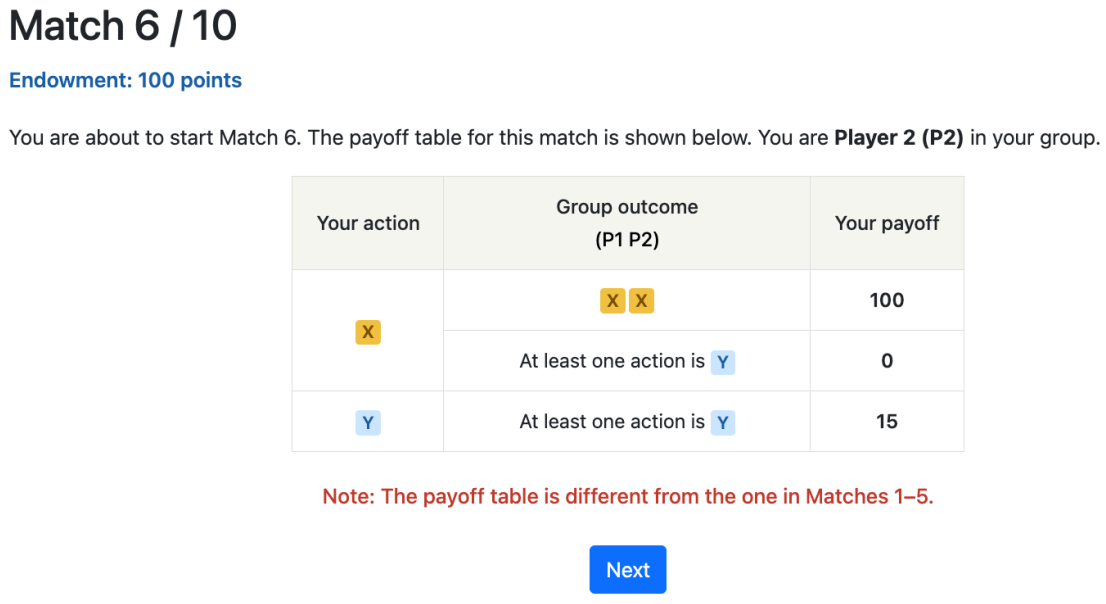


Figure C1.7.2: Welcome page for Matches 6-10 ($n = 2$ sessions)

The order of the welcome pages shown here ($u_Y = 85$ for Matches 1-5, $u_Y = 15$ for Matches 6-10) is for the $2h{+}2l$ treatment and baseline sessions. For the $2l{+}2h$ treatment and baseline

sessions, the two payoff tables will reverse the order in display ($u_Y = 15$ for Matches 1-5, $u_Y = 85$ for Matches 6-10).

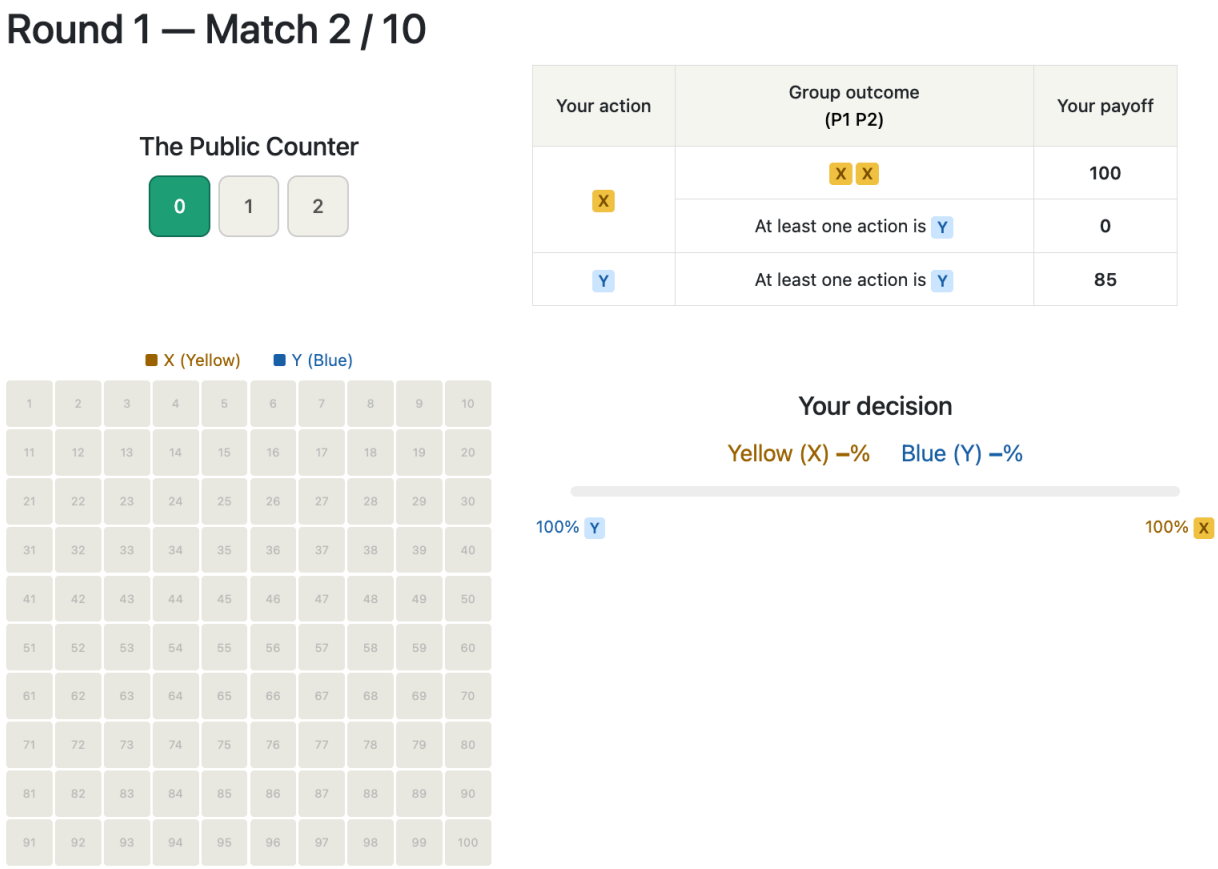


Figure C1.7.3: Decision page when Counter=0 ($n = 2$ treatment sessions)

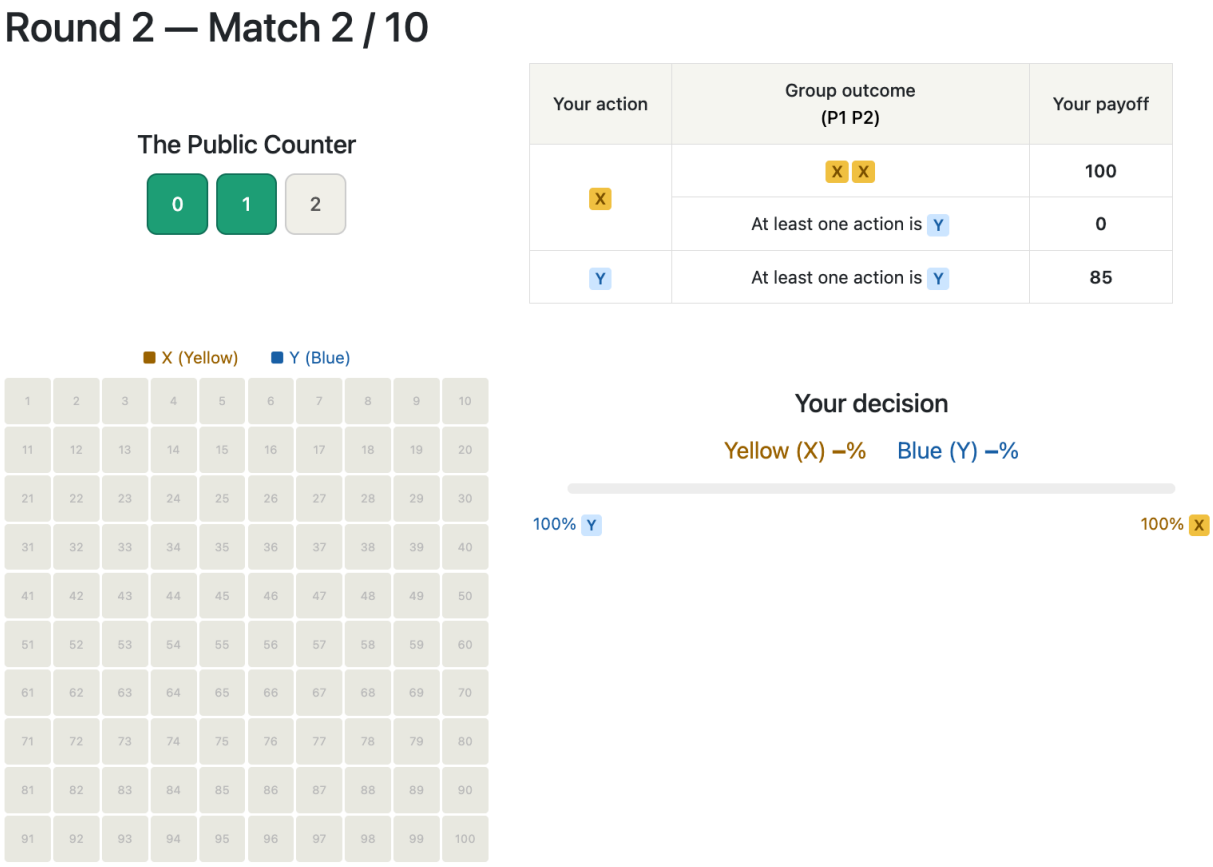


Figure C1.7.4: Decision page when Counter=1 ($n = 2$ treatment sessions)

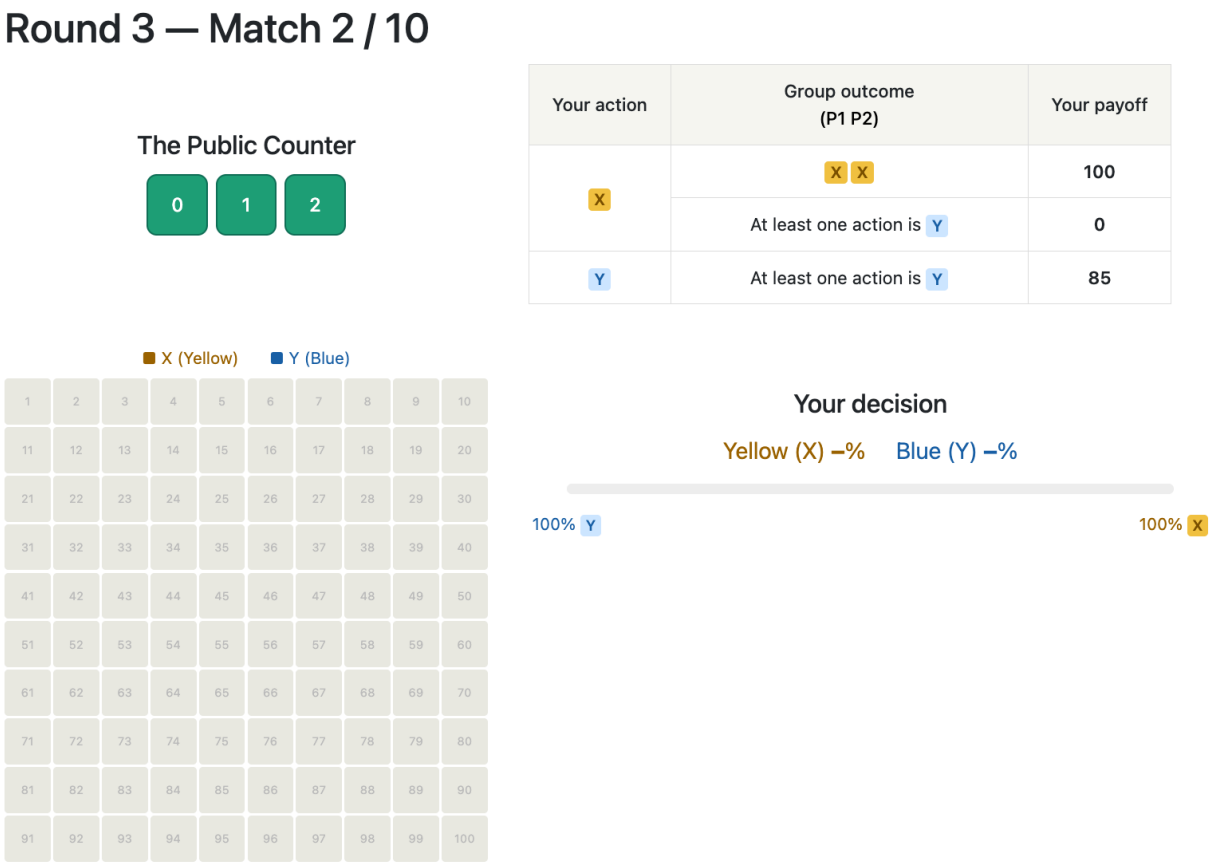


Figure C1.7.5: Decision page when Counter=2 ($n = 2$ treatment sessions)

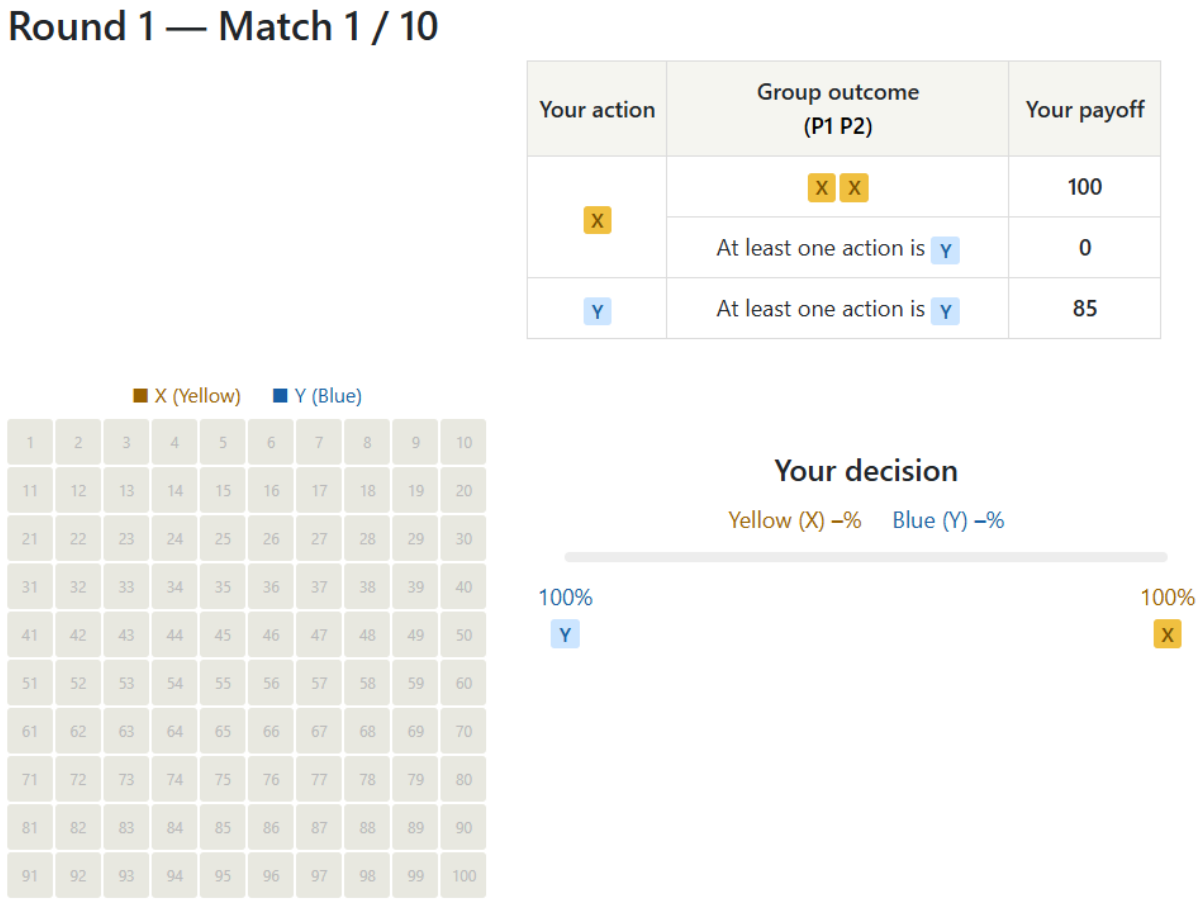


Figure C1.7.6: Decision page ($n = 2$ baseline sessions)

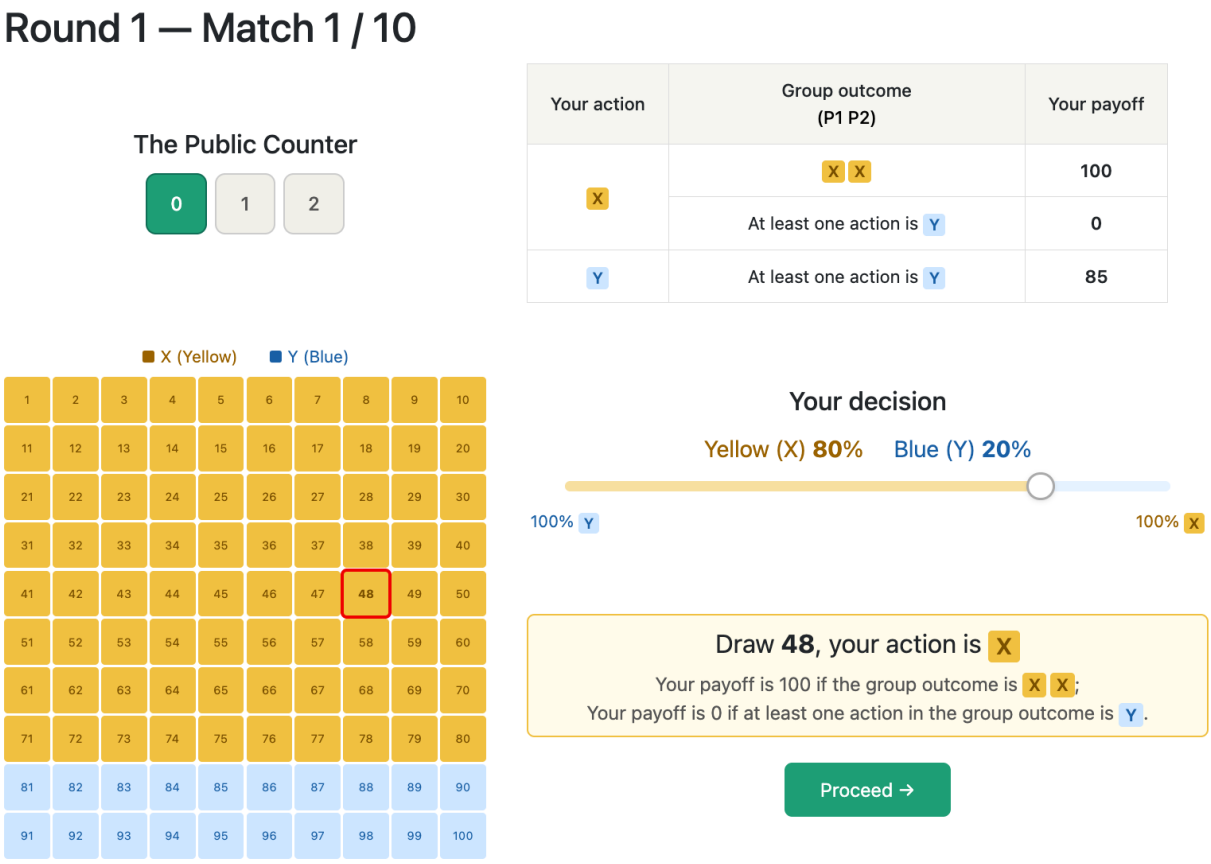


Figure C1.7.7: RevealAction page when Action X is realized ($n = 2$ treatment sessions)

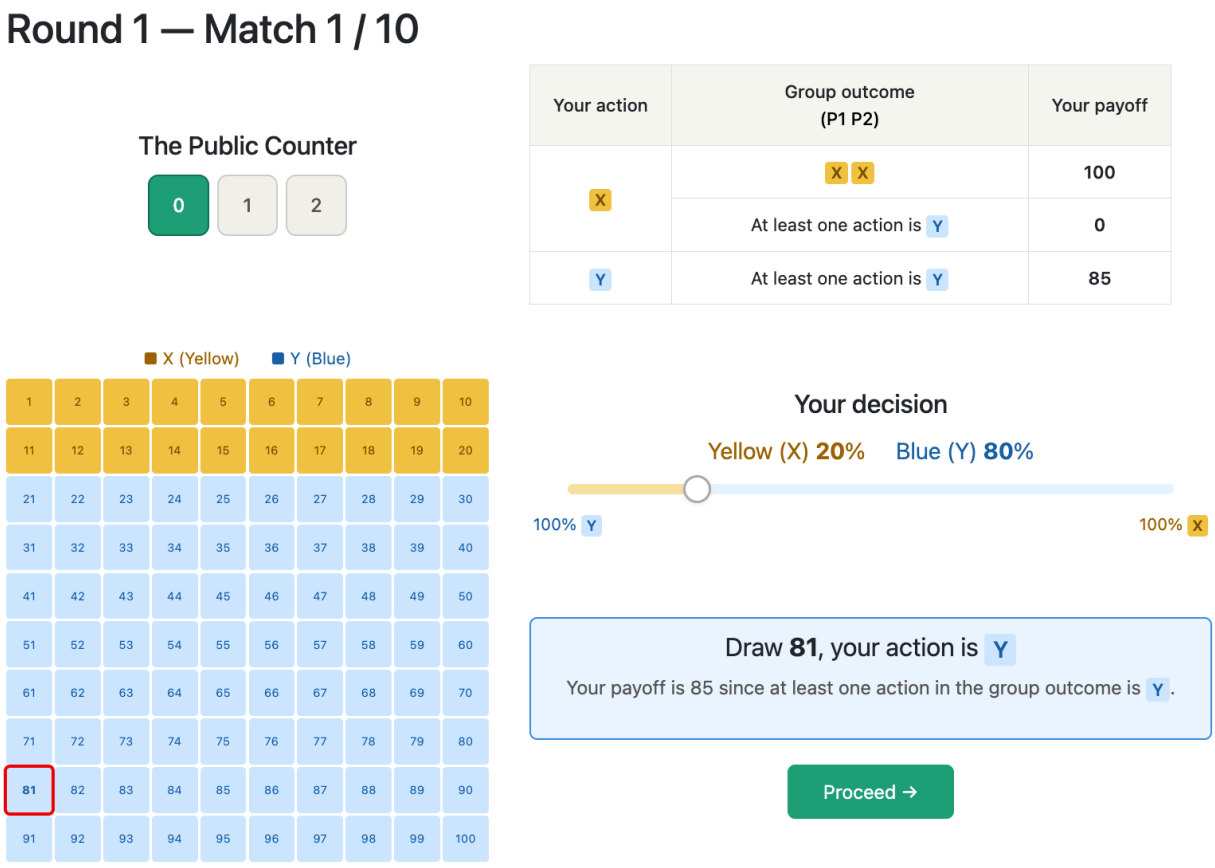


Figure C1.7.8: RevealAction page when Action Y is realized ($n = 2$ treatment sessions)

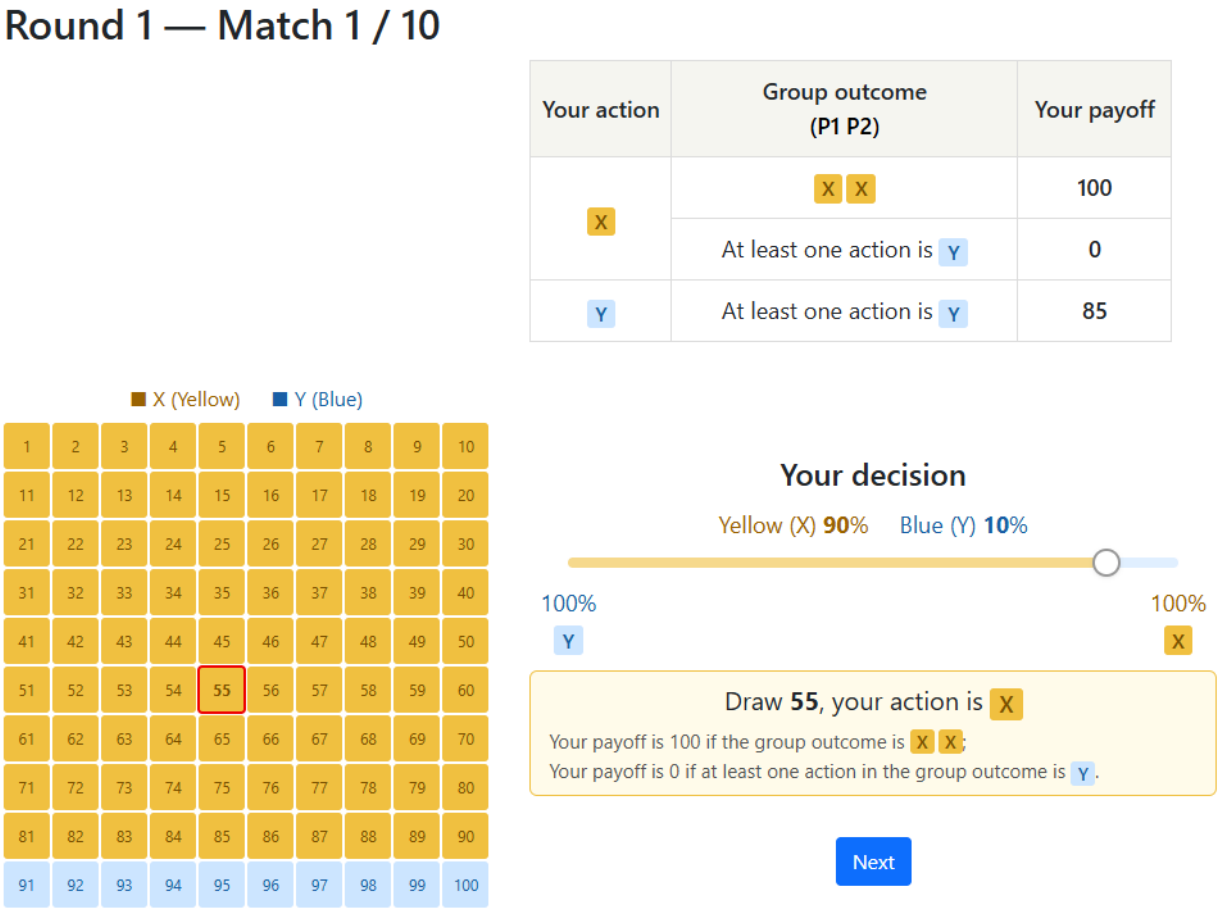


Figure C1.7.9: RevealAction page when Action X is realized ($n = 2$ baseline sessions)

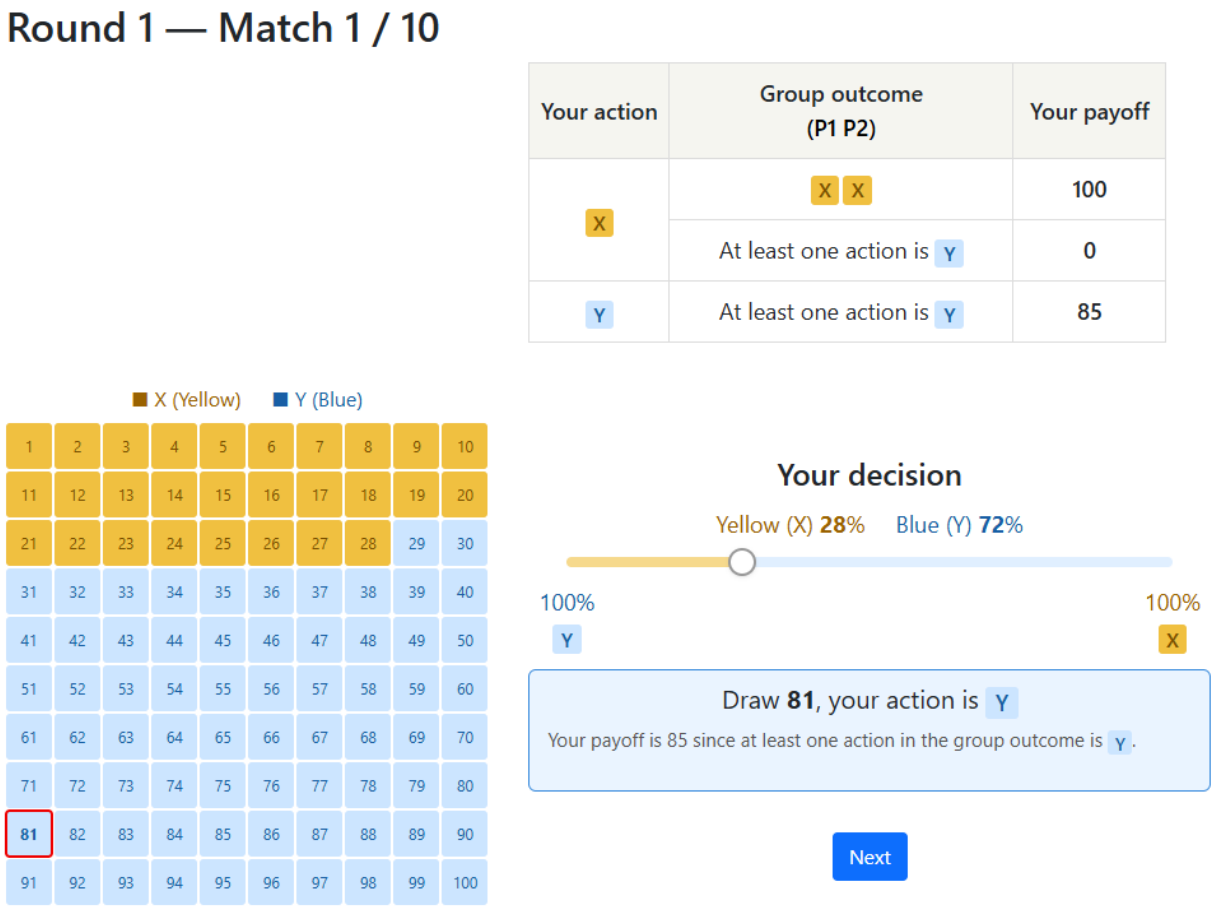


Figure C1.7.10: RevealAction page when Action Y is realized ($n = 2$ baseline sessions)

Round 3 — Match 1 / 10: Results

Endowment: 100 points

You are **Player 1 (P1)** in your group.

Round	Your probabilities		Your action	Group outcome (P1 P2)	Your payoff	Public Counter	Does the match continue?
	X	Y					
3	87%	13%	X	X X	-100	1	Ends
2	61%	39%	X	X Y	0	0	Continues
1	80%	20%	X	X X	100	1	Continues

Next

Figure C1.7.11: Result page with a penalty in the last round ($n = 2$ treatment sessions)

Round 5 — Match 4 / 10: Results

Endowment: 100 points

You are **Player 1 (P1)** in your group.

Round	Your probabilities		Your action	Group outcome (P1 P2)	Your payoff	Public Counter	Does the match continue?
	X	Y					
5	100%	0%	X	X X	100	2	Ends
4	100%	0%	X	X X	100	2	Continues
3	100%	0%	X	X X	100	1	Continues
2	87%	13%	Y	Y X	15	0	Continues
1	0%	100%	Y	Y Y	15	0	Continues

Next

Figure C1.7.12: Result page with no penalty in the last round ($n = 2$ treatment sessions)

Round 4 — Match 8 / 10: Results

Endowment: 100 points

You are **Player 2 (P2)** in your group.

Round	Your probabilities		Your action	Group outcome (P1 P2)	Your payoff	Does the match continue?
	X	Y				
4	98%	2%	X	X X	100	Ends
3	28%	72%	Y	X Y	15	Continues
2	84%	16%	X	Y X	0	Continues
1	97%	3%	X	X X	100	Continues

Next

Figure C1.7.13: Result page ($n = 2$ baseline sessions)

Match 1 Complete

Endowment	Round Payoffs	Total Earnings
100 points	-15 points	85 points

You will be randomly grouped with 1 other participant in the next match.

Next

Figure C1.7.14: Match end page

C1.8 Part 1: 10 Matches ($n = 4$ sessions)

To reduce repetition, all screenshots in this part show the Part 1 interfaces in page order, for the $n = 4$ sessions only. Explanations are provided in figure titles or between figures.

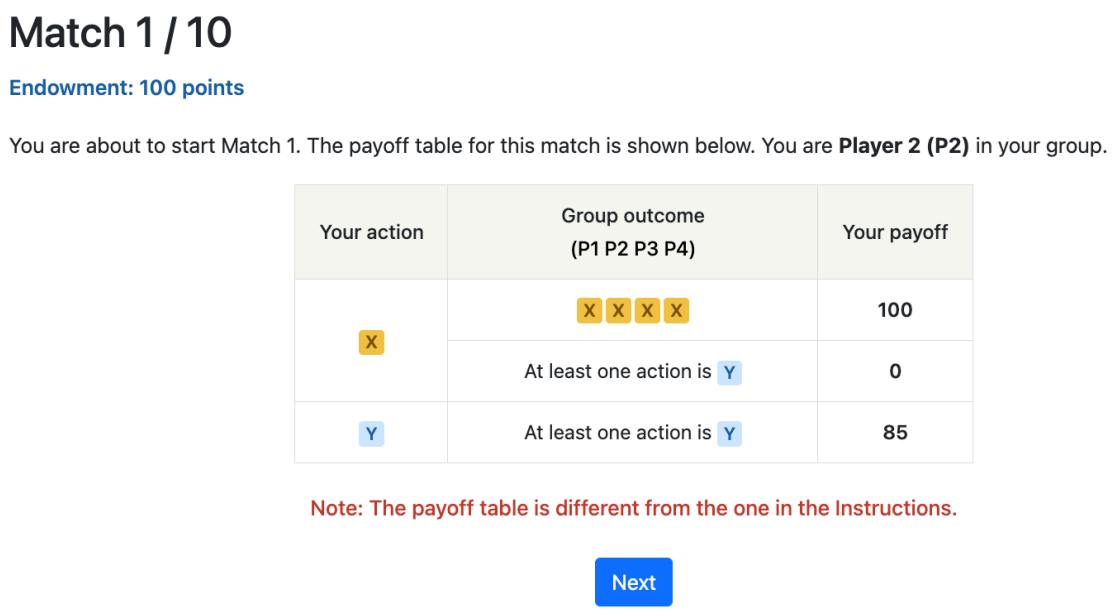


Figure C1.8.1: Welcome page for Matches 1-5 ($n = 4$ sessions)

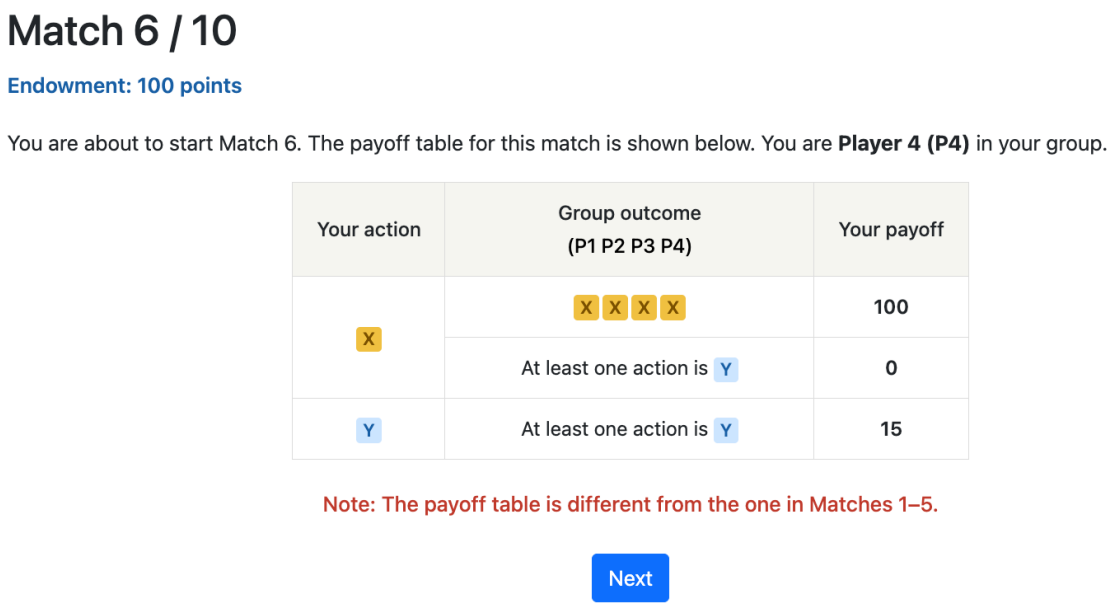


Figure C1.8.2: Welcome page for Matches 6-10 ($n = 4$ sessions)

The order of the welcome pages shown here ($u_Y = 85$ for Matches 1-5, $u_Y = 15$ for Matches

6-10) is for the 4h+4l treatment and baseline sessions. For the 4l+4h treatment and baseline sessions, the two payoff tables will reverse the order in display ($u_Y = 15$ for Matches 1-5, $u_Y = 85$ for Matches 6-10).

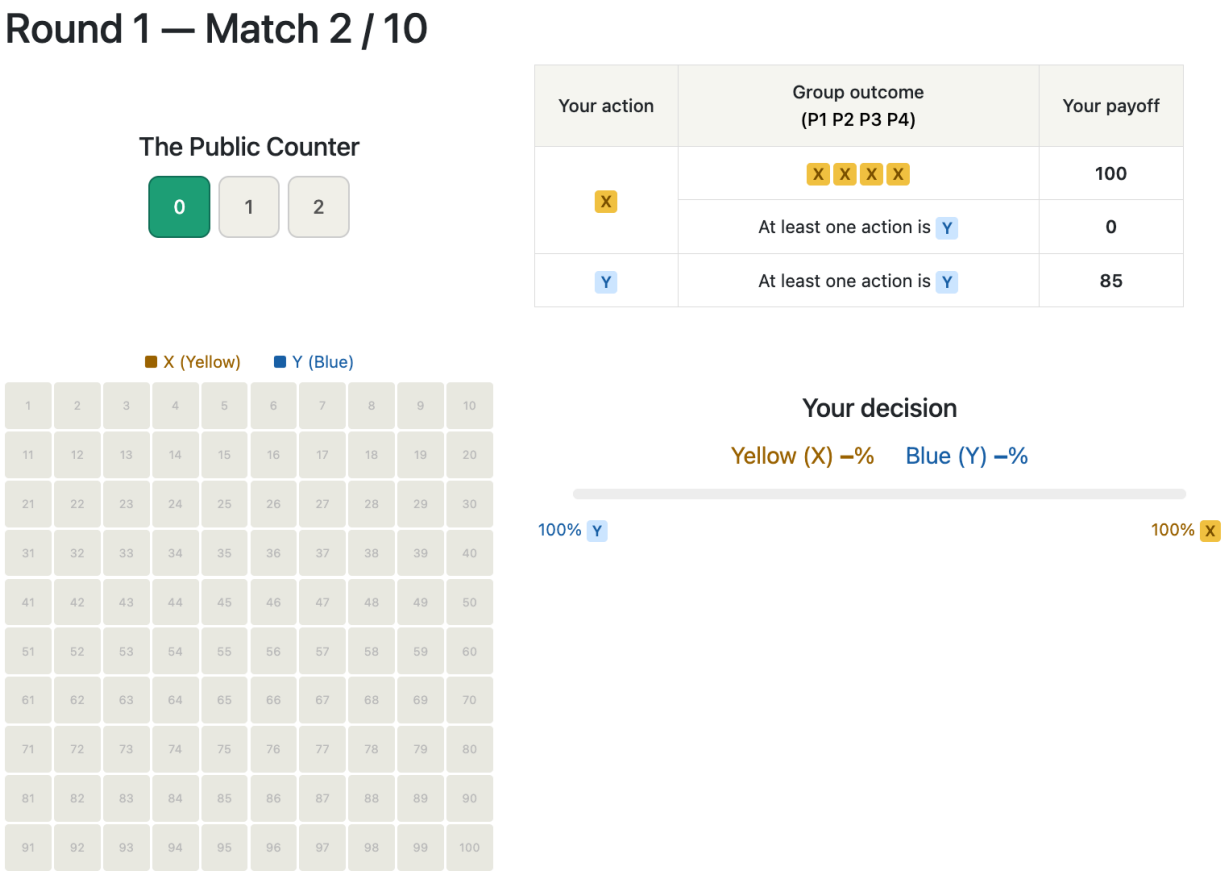


Figure C1.8.3: Decision page when Counter=0 ($n = 4$ treatment sessions)

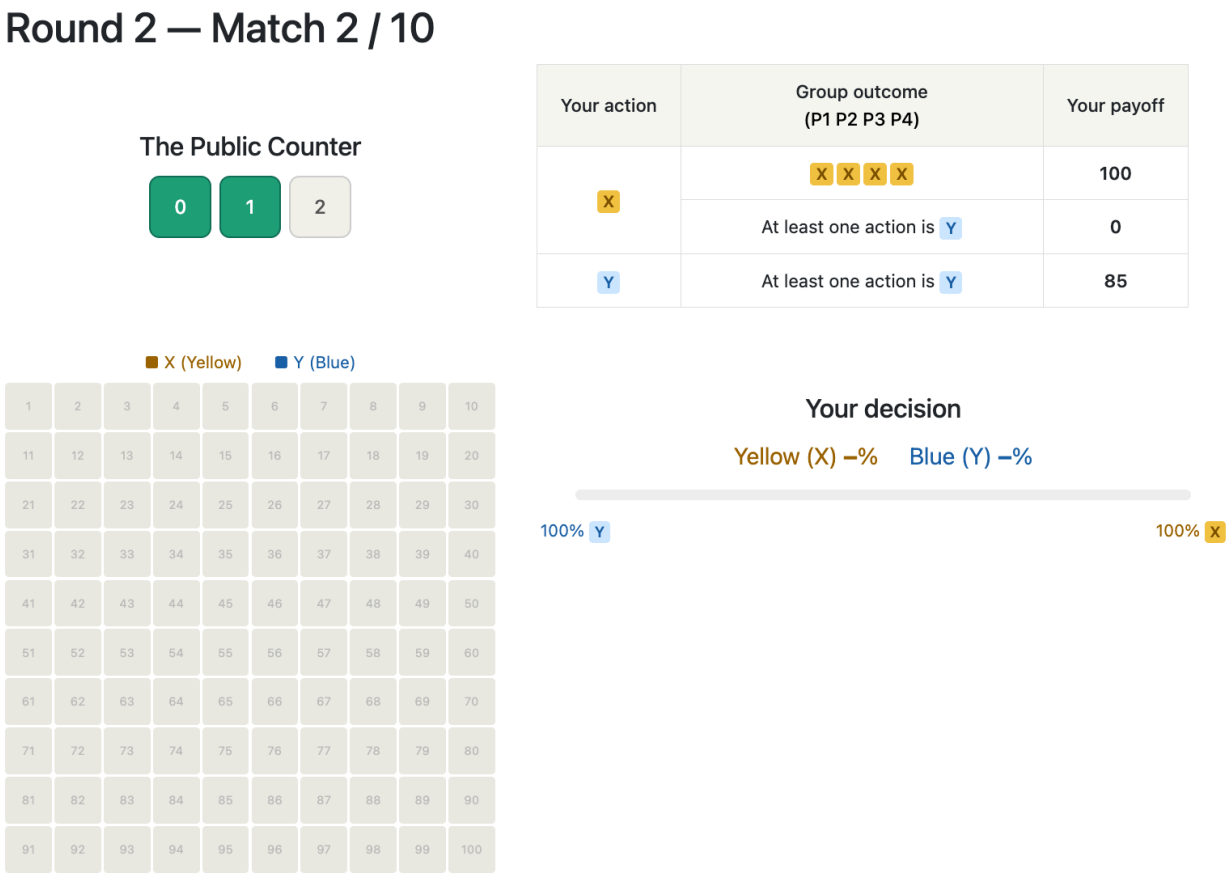


Figure C1.8.4: Decision page when Counter=1 ($n = 4$ treatment sessions)

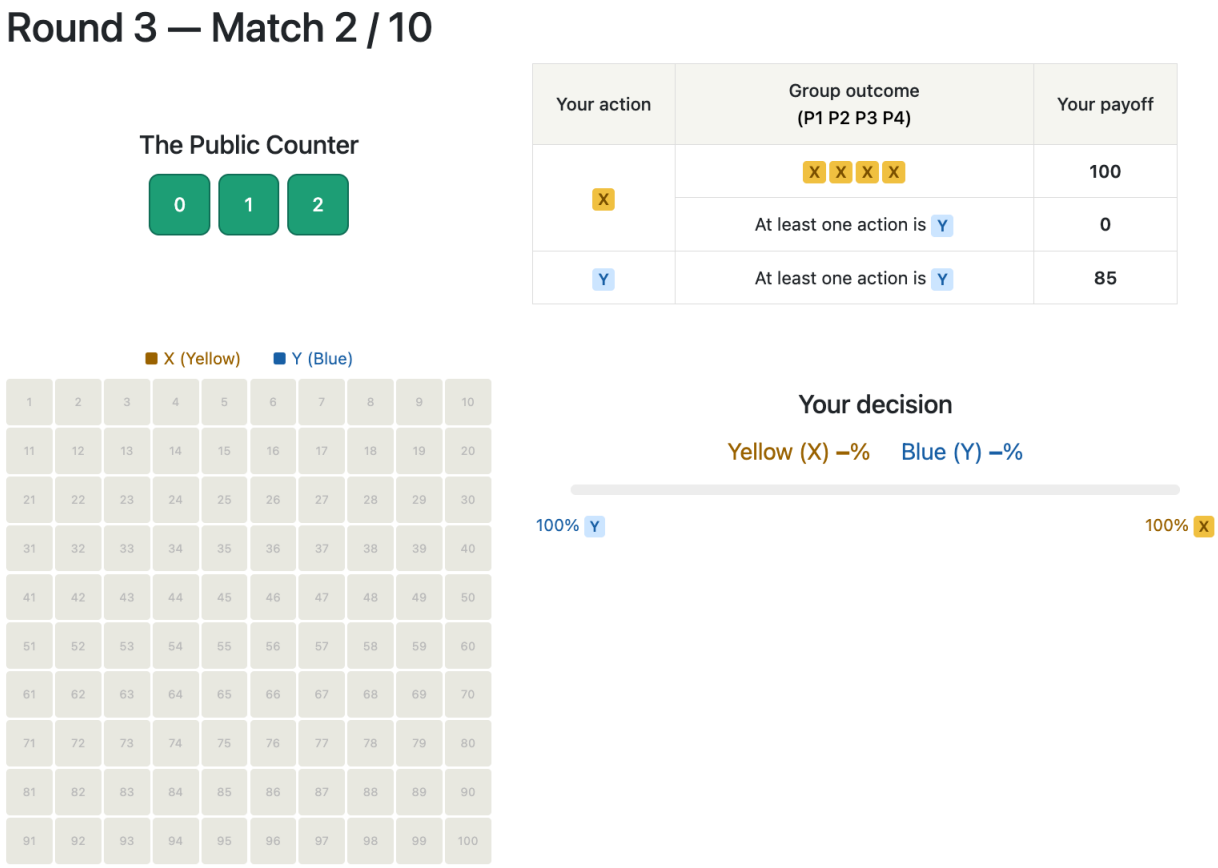


Figure C1.8.5: Decision page when Counter=2 ($n = 4$ treatment sessions)

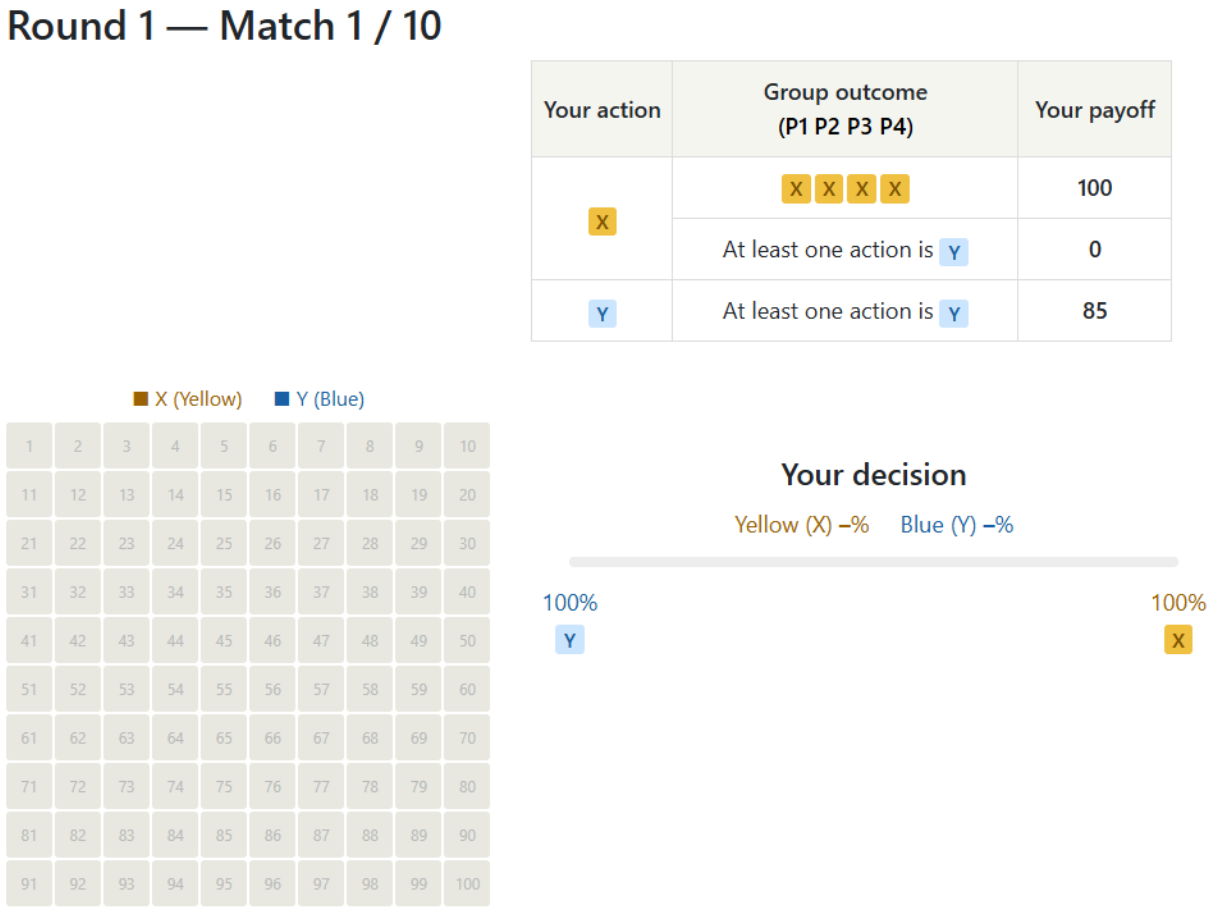


Figure C1.8.6: Decision page ($n = 4$ baseline sessions)

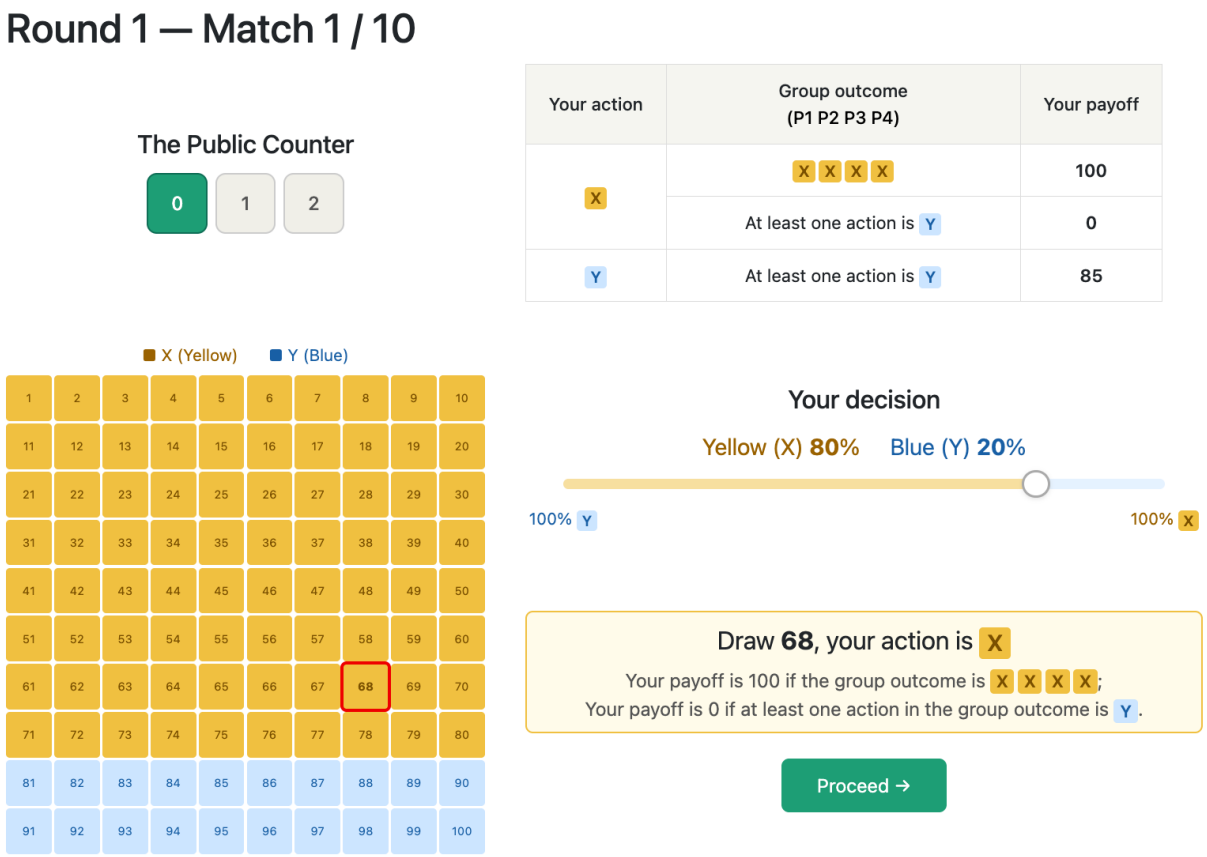


Figure C1.8.7: RevealAction page when Action X is realized ($n = 4$ treatment sessions)

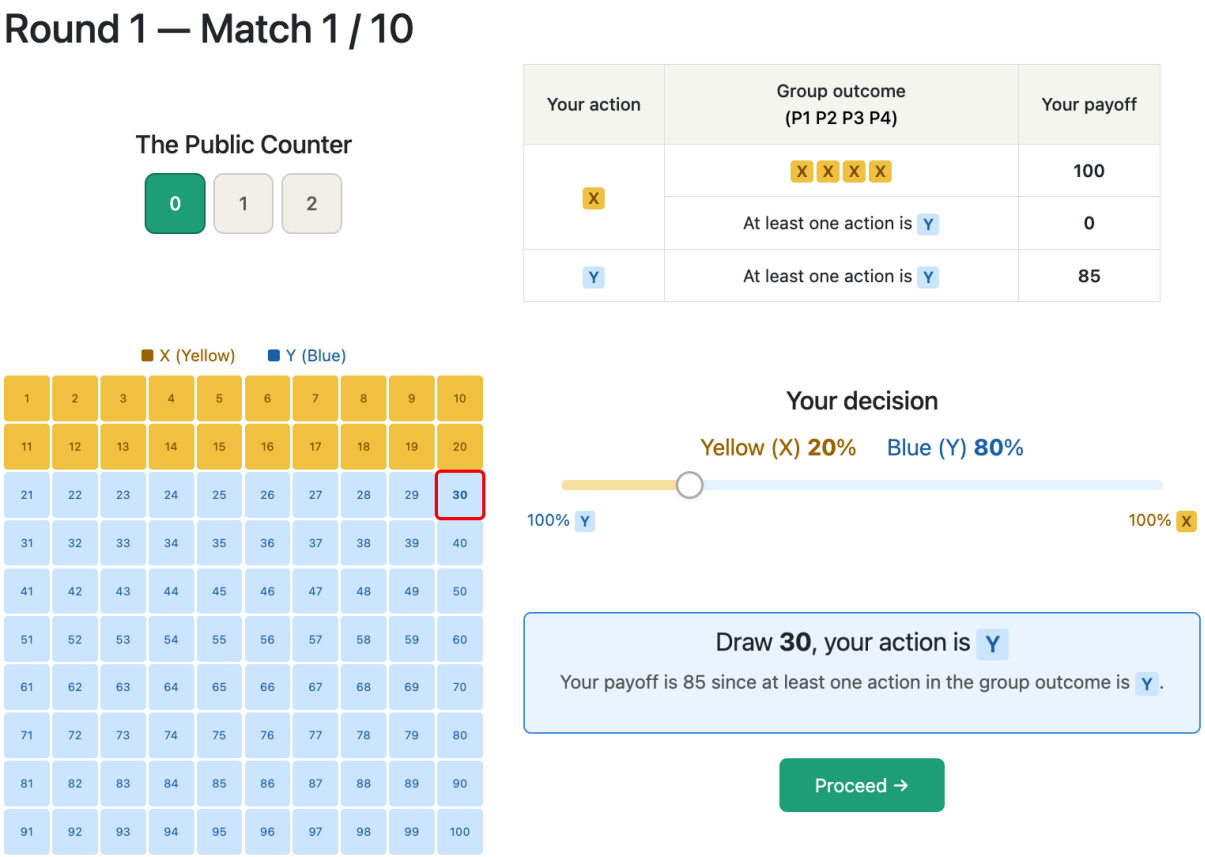


Figure C1.8.8: RevealAction page when Action Y is realized ($n = 4$ treatment sessions)

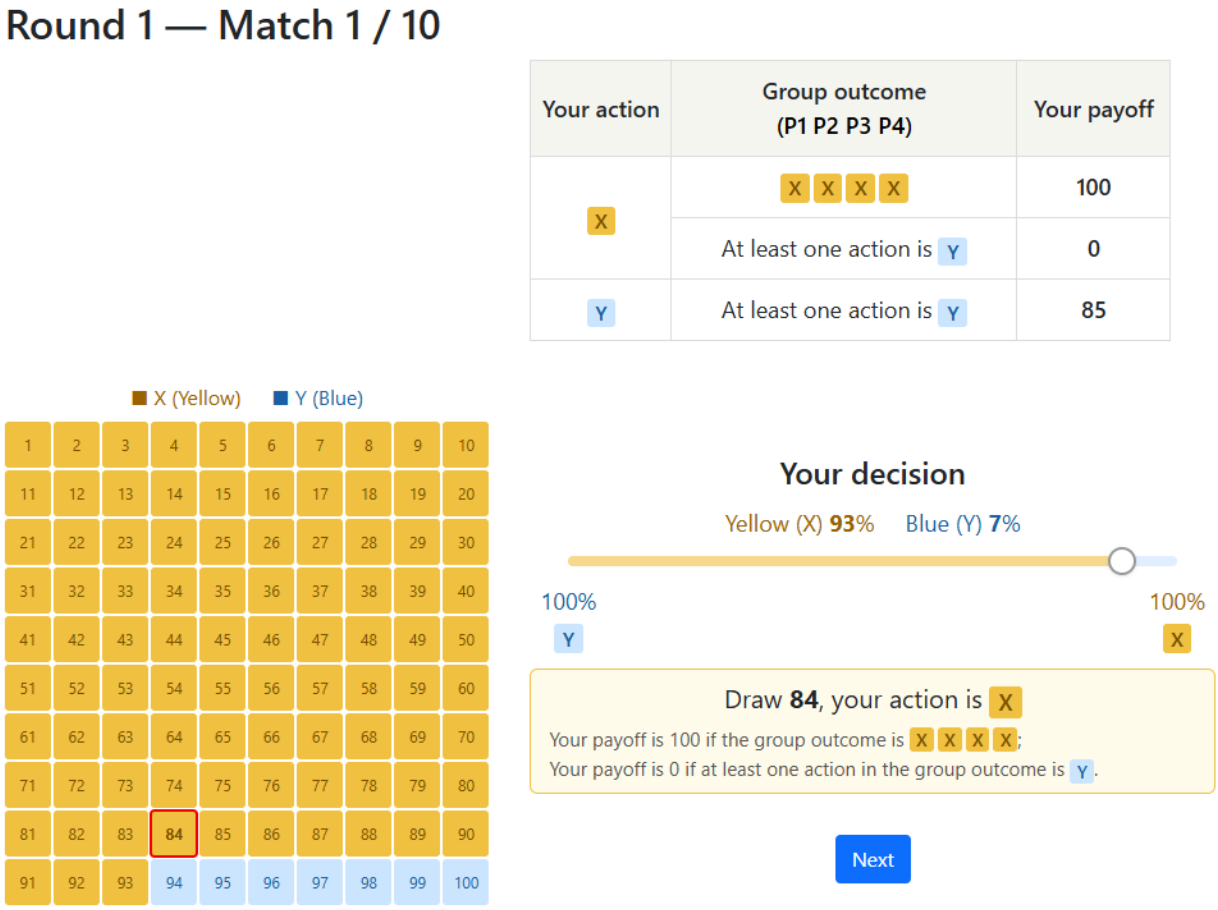


Figure C1.8.9: RevealAction page when Action X is realized ($n = 4$ baseline sessions)

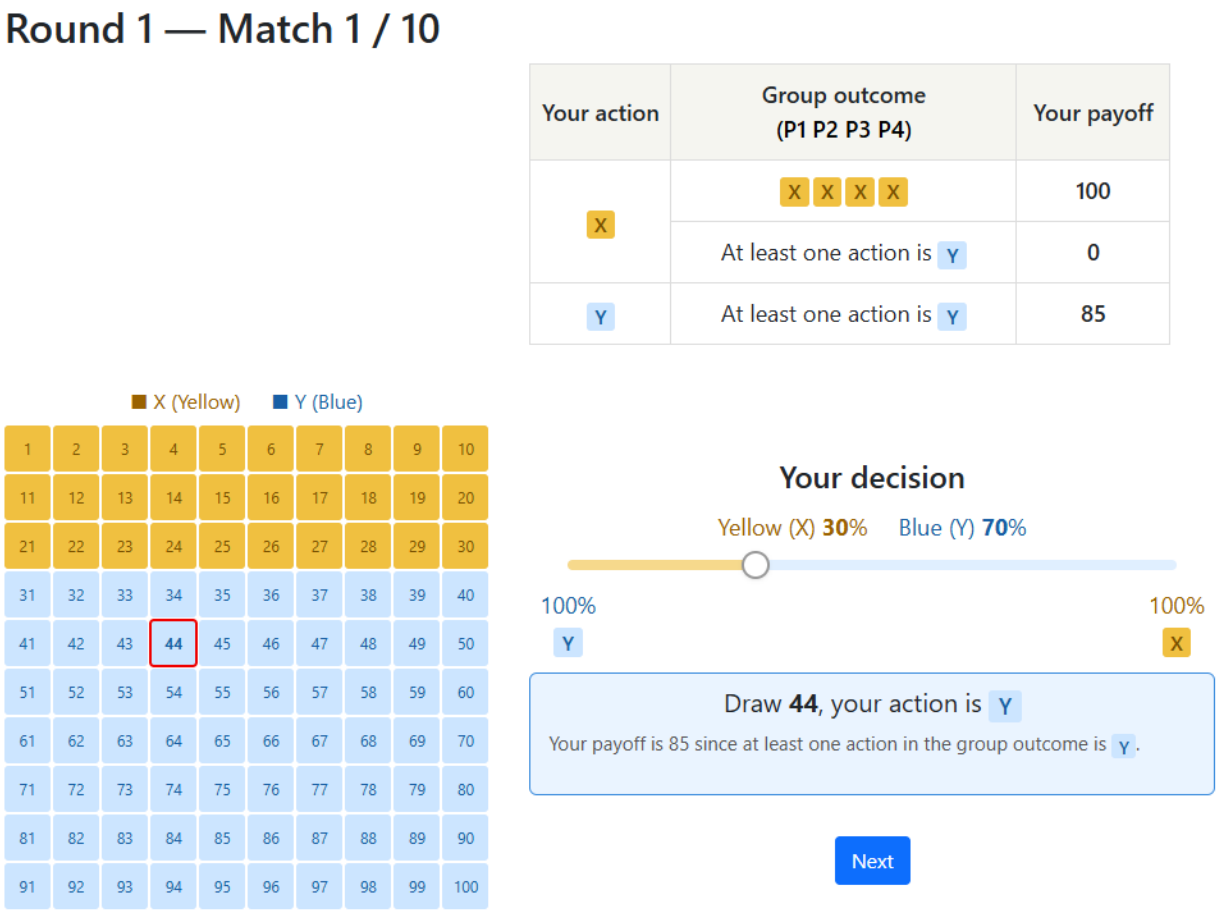


Figure C1.8.10: RevealAction page when Action Y is realized ($n = 4$ baseline sessions)

Round 3 — Match 1 / 10: Results

Endowment: 100 points

You are **Player 2 (P2)** in your group.

Round	Your probabilities		Your action	Group outcome (P1 P2 P3 P4)	Your payoff	Public Counter	Does the match continue?
	X	Y					
3	100%	0%	X	X X X Y	-100	0	Ends
2	100%	0%	X	X X X X	100	1	Continues
1	88%	12%	X	X X Y Y	0	0	Continues

Next

Figure C1.8.11: Result page with a penalty in the last round ($n = 4$ treatment sessions)

Round 5 — Match 2 / 10: Results

Endowment: 100 points

You are **Player 3 (P3)** in your group.

Round	Your probabilities		Your action	Group outcome (P1 P2 P3 P4)	Your payoff	Public Counter	Does the match continue?
	X	Y					
5	100%	0%	X	X X X X	100	2	Ends
4	100%	0%	X	X X X X	100	2	Continues
3	100%	0%	X	X X X X	100	1	Continues
2	100%	0%	X	X Y X X	0	0	Continues
1	100%	0%	X	X X X Y	0	0	Continues

Next

Figure C1.8.12: Result page with no penalty in the last round ($n = 4$ treatment sessions)

Round 4 — Match 8 / 10: Results

Endowment: 100 points

You are **Player 1 (P1)** in your group.

Round	Your probabilities		Your action	Group outcome (P1 P2 P3 P4)	Your payoff	Does the match continue?
	X	Y				
4	22%	78%	Y	Y X X X	15	Ends
3	97%	3%	X	X X X X	100	Continues
2	47%	53%	Y	Y X Y X	15	Continues
1	96%	4%	X	X X X X	100	Continues

Next

Figure C1.8.13: Result page ($n = 4$ baseline sessions)

Match 1 Complete

Endowment	Round Payoffs	Total Earnings
100 points	-100 points	0 points

You will be randomly grouped with 3 other participants in the next match.

Next

Figure C1.8.14: Match end page

C1.9 Part 2: Risk Preferences

Part 2 is identical across all sessions; therefore, only the interfaces are shown, without additional figure titles.

Part 2: Instructions

In Part 2, you will be asked to make a series of decisions. Each decision is a choice between two options -- **Option A** and **Option B**. The decisions will be organized into a table like the one below. Notice that for practice there are a total of 5 decisions (one for each rows of the table). You should think of each row as a separate decision you need to make. At this time, please make your choice for each decision row. Note that your compensation will not depend on the practice task.

Decision	Option A		Option B	Your Choice
#1	**$5.00** with 50% chance; **$0.00** with 50% chance	○ ○	**$1.00**	__
#2	**$5.00** with 50% chance; **$0.00** with 50% chance	○ ○	**$2.00**	__
#3	**$5.00** with 50% chance; **$0.00** with 50% chance	○ ○	**$3.00**	__
#4	**$5.00** with 50% chance; **$0.00** with 50% chance	○ ○	**$4.00**	__
#5	**$5.00** with 50% chance; **$0.00** with 50% chance	○ ○	**$5.00**	__

Part 2: Instructions

For Part 2, you will make 10 decisions. At the end of the experiment, one of these decisions will be chosen at random to be carried out. As an example, we will demonstrate how compensation works with the practice decisions that you just did.

The procedure involves two steps:

- First, one of the decisions will be chosen at random. For example, in the practice task you had 5 decisions, so one will be chosen randomly by the computer (with equal probability).
- Second, depending on the chosen decision in step one, a random draw might be necessary. For example, if you chose Option A over Option B, then a random draw will be made to determine the outcome of the technology. For example, in the practice task, the technology in Option A had 50% of $5.00 and 50% of $0.00.

To demonstrate the compensation procedure, we will perform the two steps using the practice task.

Decision	Option A	Option B	Your Choice
#1	**$5.00** with 50% chance; **$0.00** with 50% chance	$1.00	A
#2	**$5.00** with 50% chance; **$0.00** with 50% chance	$2.00	A
#3	**$5.00** with 50% chance; **$0.00** with 50% chance	$3.00	B
#4	**$5.00** with 50% chance; **$0.00** with 50% chance	$4.00	B
#5	**$5.00** with 50% chance; **$0.00** with 50% chance	$5.00	B

1. The randomly chosen decision is Decision #2

2. The random realization of Option A was $0.00.

Therefore, your payoff is $0.00.

To gain further understanding of the compensation procedure, you may try again.

Try Again

In the actual experiment, this procedure will be carried out **only once**. If you are ready to begin, please click **Next**.

Next

Part 2: Decisions

Please make a choice for **each of the 10 decisions**.

Decision	Option A		Option B	Your Choice
#1	**$5.00** with 50% chance; **$0.00** with 50% chance	○ ○	**$0.50**	__
#2	**$5.00** with 50% chance; **$0.00** with 50% chance	○ ○	**$1.00**	__
#3	**$5.00** with 50% chance; **$0.00** with 50% chance	○ ○	**$1.50**	__
#4	**$5.00** with 50% chance; **$0.00** with 50% chance	○ ○	**$2.00**	__
#5	**$5.00** with 50% chance; **$0.00** with 50% chance	○ ○	**$2.50**	__
#6	**$5.00** with 50% chance; **$0.00** with 50% chance	○ ○	**$3.00**	__
#7	**$5.00** with 50% chance; **$0.00** with 50% chance	○ ○	**$3.50**	__
#8	**$5.00** with 50% chance; **$0.00** with 50% chance	○ ○	**$4.00**	__
#9	**$5.00** with 50% chance; **$0.00** with 50% chance	○ ○	**$4.50**	__
#10	**$5.00** with 50% chance; **$0.00** with 50% chance	○ ○	**$5.00**	__

C1.10 Part 3: Social Value Orientations

Part 3 is identical across all sessions; therefore, only the interfaces are shown, without additional figure titles.

Part 3 : Instructions

In this part of the study, you will be randomly paired with another person, whom we will refer to as the **other**. You will not know who the other person is, nor will the other person be informed about your identity. You will make a series of choices among several alternative allocations of points. These points will be converted into Dollars at a rate of **50 points = $1.00**. Note that this rate is different from the one in Part 1.

You will be making a series of decisions about allocating points between you and this other person. For each of the questions, please indicate the distribution you prefer most by selecting the corresponding button in the middle row. You can only make one choice for each question. There are no right or wrong answers, this is all about personal preference.

In the example below, a person chose the allocation giving 50 points to herself, and 40 points to the unknown other person. In terms of US dollars, this yields an allocation of 50 / 50 = $1.00 for the person making the choice and 40 / 50 = $0.80 for the unknown other.

You Receive	30	35	40	45	50	55	60	65	70
	○	○	○	○	◉	○	○	○	○
Other Receives	80	70	60	50	40	30	20	10	0
									Submit

As you can see, your choices influence both the number of points you receive, as well as the number of points the other person receives.

After you have made all your choices, one of the allocation choices will be randomly selected by the computer. **For this choice, the computer will randomly assign one person from your group (you or the other) the role of "Receiver" and the other the role of the "Sender". The allocation choice made by the Sender will be enforced.** This allocation will be paid to both the Sender and the Receiver.

If you have any questions, please raise your hand.

Next

Part 3 : Decision 1 of 6

Choose a distribution between yourself and the other individual

You Receive	85	85	85	85	85	85	85	85	85
	○	○	○	○	○	○	○	○	○
Other Receives	85	76	68	59	50	41	33	24	15
									Submit

Part 3 : Decision 2 of 6

Choose a distribution between yourself and the other individual

You Receive	85	87	89	91	93	94	96	98	100
	○	○	○	○	○	○	○	○	○
Other Receives	15	19	24	28	33	37	41	46	50

Submit

Part 3 : Decision 3 of 6

Choose a distribution between yourself and the other individual

You Receive	50	54	59	63	68	72	76	81	85
	○	○	○	○	○	○	○	○	○
Other Receives	100	98	96	94	93	91	89	87	85

Submit

Part 3 : Decision 4 of 6

Choose a distribution between yourself and the other individual

You Receive	50	54	59	63	68	72	76	81	85
	○	○	○	○	○	○	○	○	○
Other Receives	100	89	79	68	58	47	36	26	15

Submit

Part 3 : Decision 5 of 6

Choose a distribution between yourself and the other individual

You Receive	100	94	88	81	75	69	63	56	50
	○	○	○	○	○	○	○	○	○
Other Receives	50	56	63	69	75	81	88	94	100

Submit

Part 3 : Decision 6 of 6

Choose a distribution between yourself and the other individual

You Receive	100	98	96	94	93	91	89	87	85
	○	○	○	○	○	○	○	○	○
Other Receives	50	54	59	63	68	72	76	81	85

Submit

C1.11 Survey

Questions 1 to 4 and 7 are identical across sessions. Question 5 differs between treatment and baseline sessions, while Question 6 differs between the $n = 2$ and $n = 4$ sessions.

Please complete this short survey while we prepare your payment information.

1. What's your gender?
   - Female
   - Male
   - Other
   - Prefer not to say

2. Did you complete high school in an English-speaking country or region?
   - Yes
   - No
   - Prefer not to say

3. Which of the following best describes your major?
   - Agriculture
   - Business and Economics
   - Education
   - Engineering
   - Health and Human Sciences
   - Liberal Arts and Humanities
   - Other Social Sciences
   - Science and Mathematics
   - Others

4. Using 1-2 sentences, briefly describe your strategies of assigning probabilities to X and Y.

   Please describe your strategy...

5. *(For treatment sessions)* Within a match, did your decision of assigning probabilities to X and Y in a round depend on the current counter, on group outcomes in previous rounds, or both?

- The current counter only
- Group outcomes in previous rounds only
- Both the current counter and group outcomes in previous rounds

5. *(For baseline sessions)* Within a match, did your decision of assigning probabilities to X and Y in a round depend on the group outcomes in previous rounds?

- Yes
- No

6. Within Matches 1-5 and 6-10, was your strategy different between the initial matches and the later matches?

$n = 2$ *sessions*

Payoff table for Matches 1-5

Your action	Group outcome (P1 P2)	Your payoff
X	X X	100
	At least one action is Y	0
Y	At least one action is Y	85

Payoff table for Matches 6-10

Your action	Group outcome (P1 P2)	Your payoff
X	X X	100
	At least one action is Y	0
Y	At least one action is Y	15

$n = 4$ *sessions*

Payoff table for Matches 1-5

Your action	Group outcome (P1 P2 P3 P4)	Your payoff
X	X X X X	100
	At least one action is Y	0
Y	At least one action is Y	85

Payoff table for Matches 6-10

Your action	Group outcome (P1 P2 P3 P4)	Your payoff
X	X X X X	100
	At least one action is Y	0
Y	At least one action is Y	15

- Same within Matches 1-5; Same within Matches 6-10
- Different within Matches 1-5; Same within Matches 6-10
- Same within Matches 1-5; Different within Matches 6-10
- Different within Matches 1-5; Different within Matches 6-10

The payoff tables shown above correspond to the high-first treatment and baseline sessions ($2h+2l$ and $4h+4l$). In the low-first treatment and baseline sessions ($2l+2h$ and $4l+4h$), the two payoff tables are displayed in reverse order.

7. Were the strategies you use for Matches 6-10 influenced by the ones you use for Matches 1-5?

- Yes
- No

C1.12 Final Payment Page

Payment Summary

All 10 matches from Part 1 are paid. Exchange rate: **400 points = $1.00**.

Match	1	2	3	4	5	6	7	8	9	10	Total
Points	355	780	270	525	695	220	220	160	115	130	**3470**

Comprehension quiz (0/6 correct)	$0.00
Part 1 (3470 points)	$8.68
Part 2	$5.00
Part 3	$1.00
Final payment	**$14.68**

Please remain seated. The experimenter will come to you shortly.

Appendix C2: Continuation Draws and Power Analysis

C2.1 Continuation Draws

Table C2.1: Continuation Draws and Realized Match Lengths

Match	Draws (Arrow landing positions)	Rounds
	Sequence #1	**Total: 48**
1	[51, 1, 95]	3
2	[20, 64, 74, 23, 20, 28, 57, 87]	8
3	[27, 98]	2
4	[70, 79, 7, 78, 88]	5
5	[37, 69, 41, 39, 60, 43, 85]	7
6	[25, 10, 48, 30, 62, 41, 2, 93]	8
7	[9, 77, 60, 7, 25, 31, 8, 91]	8
8	[51, 1, 73, 87]	4
9	[95]	1
10	[14, 90]	2
	Sequence #2	**Total: 45**
1	[49, 67, 88]	3
2	[48, 40, 17, 21, 93]	5
3	[25, 89]	2
4	[95]	1
5	[67, 8, 79, 11, 60, 33, 43, 5, 90]	9
6	[70, 42, 83]	3
7	[72, 7, 49, 23, 89]	5
8	[78, 74, 10, 58, 55, 96]	6
9	[14, 39, 62, 27, 75, 13, 48, 71, 20, 92]	10
10	[94]	1
	Sequence #3	**Total: 50**
1	[68, 14, 85]	3
2	[74, 19, 19, 1, 35, 44, 30, 10, 64, 97]	10
3	[78, 3, 46, 93]	4
4	[6, 74, 17, 44, 14, 89]	6
5	[66, 81]	2
6	[50, 4, 91]	3
7	[80, 6, 22, 87]	4
8	[46, 69, 90]	3
9	[18, 43, 66, 12, 68, 82]	6
10	[55, 55, 50, 25, 28, 33, 44, 24, 84]	9

Table C2.1 reports the predrawn continuation draws for each round of the 10 matches in Part 1. Each draw is an integer from the discrete uniform distribution on $\{1, \ldots, 100\}$. A

draw at or below 80 indicates that the match continues to another round, whereas a draw above 80 terminates the match, corresponding to a continuation probability of $\delta = 0.8$. The continuation draw is revealed at the end of every round, so the match ends immediately following the first draw above 80. Accordingly, the final entry in each row of Table C2.1 is the terminating draw, and the number of draws in that row equals the realized length of the match. For example, in Match 2 of Sequence #1, the first seven draws are at or below 80 and the eighth draw is 87; the match therefore ends after Round 8.

The random draws are presented to subjects using a spinner with an 80% white continuation region and a 20% gray stopping region. A draw at or below 80 corresponds to the arrow landing in the white region, while a draw above 80 corresponds to the arrow landing in the gray region. Subjects observe the spinner realization after each round and learn at that point whether the match continues or ends. The three sequences contain 48, 45, and 50 rounds in total, respectively, across their 10 matches.

Each of the three sequences was used in four treatment sessions, once for each treatment order: $2l + 2h$, $2h + 2l$, $4l + 4h$, and $4h + 4l$. Thus, the three sequences generated the 12 treatment sessions. The $n = 2$ sessions contained 10-12 subjects, while each $n = 4$ session contained 16 subjects, for a total treatment sample of 164 subjects.

C2.2 Power Analysis

I conducted an ex ante power analysis to assess the precision of alternative sample sizes for the main experiment. Because the group-size treatment is assigned at the session level, the comparison between $n = 2$ and $n = 4$ is the most demanding of the treatment comparisons and therefore serves as the power-driving test. The outcome used for this analytical power calculation is the subject-level mean probability assigned to action X, denoted by $\bar{p}_i(X) \in [0, 1]$.

The experiment is implemented in balanced blocks based on the predetermined sequences of match length.[18] For each sequence, I conduct four sessions corresponding to the four treatment combinations: $n = 2$ with $u_Y = 0.15$ first, $n = 2$ with $u_Y = 0.85$ first, $n = 4$ with $u_Y = 0.15$ first, and $n = 4$ with $u_Y = 0.85$ first. Each $n = 2$ session contains 12 subjects, and each $n = 4$ session contains 16 subjects. Thus, each complete block contains four sessions and $12 + 12 + 16 + 16 = 56$ subjects. Let S denote the number of complete sequence blocks. Since each block contains two $n = 2$ sessions and two $n = 4$ sessions, there are $2S$ sessions in each group-size treatment. Table C2.2 summarizes the assumptions.

Given the experimental budget constraint, I report minimum detectable effects (MDEs)

[18] See Appendix C2 for predrawn sequences.

Table C2.2: Power Analysis: Assumptions

Quantity	Value	Notes
Primary outcome	$\bar{p}_i(X)$	Subject-level mean probability assigned to X
Subject-level SD, σ	0.25	Baseline assumption
	0.20-0.30	Alternatives
Session-level ICC, ρ	0.05	Baseline assumption
	0.10	Conservative case
Subjects per $n = 2$ session	12	
Subjects per $n = 4$ session	16	
Sessions per sequence block	4	2 sessions for each group-size treatment
Subjects per sequence block	56	
Significance level α	0.05	One-sided
Target power	0.80	
treatment comparison		$n = 2$ vs. $n = 4$

rather than assuming a particular treatment-effect magnitude. The calculations use 80% power and a one-sided significance level of $\alpha = 0.05$, consistent with the directional prediction that coordination on X is higher in smaller groups. I use $\sigma = 0.25$ as the baseline standard deviation of the subject-level mean probability assigned to X and report sensitivity to $\sigma \in \{0.20, 0.25, 0.30\}$. To account for dependence among subjects within the same session, I consider a session-level intraclass correlation of $\rho = 0.05$ as the baseline case and $\rho = 0.10$ as a more conservative case. For this analytical approximation, I apply the same value of ρ to the two group-size treatments.

Assume that the subject-level mean probability assigned to X has variance

$$\mathrm{Var}(\bar{p}_i(X)) = \sigma^2.$$

Let ρ denote the intraclass correlation (ICC) between the subject-level mean probabilities of any two subjects i and j in the same session. Thus,

$$\mathrm{Corr}(\bar{p}_i(X), \bar{p}_j(X)) = \rho,$$

which implies

$$\mathrm{Cov}(\bar{p}_i(X), \bar{p}_j(X)) = \rho\sigma^2.$$

The temporary groups formed within a session are not treated as independent clusters because group membership changes across matches. Dependence induced by repeated interaction and rematching is instead summarized by the session-level ICC ρ. For a session s

containing m subjects, define the session-level mean as

$$\bar{p}_s(X) = \frac{1}{m}\sum_{i=1}^{m}\bar{p}_i(X).$$

The variance of the session mean is therefore

$$\begin{aligned}
\mathrm{Var}(\bar{p}_s(X)) &= \mathrm{Var}\left(\frac{1}{m}\sum_{i=1}^{m}\bar{p}_i(X)\right) \\
&= \frac{1}{m^2}\left[\sum_{i=1}^{m}\mathrm{Var}(\bar{p}_i(X)) + \sum_{i\neq j}\mathrm{Cov}(\bar{p}_i(X),\bar{p}_j(X))\right] \\
&= \frac{1}{m^2}\left[m\sigma^2 + m(m-1)\rho\sigma^2\right] \\
&= \frac{\sigma^2}{m}\left[1+(m-1)\rho\right] \\
&= \sigma^2\left[\rho + \frac{1-\rho}{m}\right].
\end{aligned}$$

Here, ρ captures the component of subject-level variance associated with variation shared by subjects within the same session, while $1-\rho$ captures subject-specific variation, which is reduced by a factor of m when averaged across subjects.

In the $n = 2$ treatment, each session contains 12 subjects. Hence, the variance of an $n = 2$ session mean is

$$V_2 = \sigma^2\left[\rho + \frac{1-\rho}{12}\right].$$

In the $n = 4$ treatment, each session contains 16 subjects. Hence, the variance of an $n = 4$ session mean is

$$V_4 = \sigma^2\left[\rho + \frac{1-\rho}{16}\right].$$

With S complete sequence blocks, there are $2S$ sessions in each group-size treatment. The variance of the mean outcome in each treatment is therefore

$$\mathrm{Var}(\bar{p}_{n=2}(X)) = \frac{V_2}{2S}, \qquad \mathrm{Var}(\bar{p}_{n=4}(X)) = \frac{V_4}{2S}.$$

Since the $n = 2$ and $n = 4$ sessions are independent, the variance of the difference in treatment means is

$$\mathrm{Var}\left(\bar{p}_{n=2}(X) - \bar{p}_{n=4}(X)\right) = \frac{V_2}{2S} + \frac{V_4}{2S}.$$

The standard error of the group-size treatment difference is therefore

$$se = \sqrt{\frac{V_2}{2S} + \frac{V_4}{2S}}.$$

Because the two group-size treatments contain different numbers of subjects per session and therefore have different session-mean variances, I use the Welch-Satterthwaite approximation for the degrees of freedom:

$$\nu = \frac{\left(\frac{V_2}{2S} + \frac{V_4}{2S}\right)^2}{\frac{(V_2/(2S))^2}{2S-1} + \frac{(V_4/(2S))^2}{2S-1}}.$$

Let $\Delta^{\min}$ denote the minimum detectable group-size effect. Its corresponding noncentrality parameter is

$$\lambda = \frac{\Delta^{\min}}{se}.$$

For a one-sided test with significance level $\alpha = 0.05$, $\Delta^{\min}$ is chosen such that

$$\Pr\left(T_{\nu,\lambda} > t_{0.95,\nu}\right) = 0.80,$$

where $T_{\nu,\lambda}$ denotes a noncentral t random variable with ν degrees of freedom and noncentrality parameter λ. For example, under the baseline assumptions $\sigma = 0.25$ and $\rho = 0.05$, with three complete sequence blocks ($S = 3$),

$$V_2 = 0.25^2\left[0.05 + \frac{0.95}{12}\right] = 0.008073,$$
$$V_4 = 0.25^2\left[0.05 + \frac{0.95}{16}\right] = 0.006836.$$

Since $S = 3$ implies six sessions in each group-size treatment,

$$se = \sqrt{\frac{0.008073}{6} + \frac{0.006836}{6}} = 0.04985.$$

The Welch-Satterthwaite degrees of freedom are

$$\nu = \frac{(0.008073/6 + 0.006836/6)^2}{\frac{(0.008073/6)^2}{5} + \frac{(0.006836/6)^2}{5}} \approx 9.93.$$

For a one-sided 5% significance level,

$$t_{0.95,9.93} \approx 1.814.$$

Solving

$$\Pr\left(T_{9.93,\lambda} > 1.814\right) = 0.80$$

gives

$$\lambda \approx 2.675.$$

Therefore,

$$\Delta^{\min} = \lambda \times se = 2.675 \times 0.04985 \approx 0.1333 = 13.3 \text{ percentage points.}$$

Thus, under the baseline variance and ICC assumptions, three sequence blocks, corresponding to 12 sessions and 168 subjects, provide 80% power to detect a group-size difference of approximately 13.3 percentage points, as shown in Table C2.3.

Table C2.3: Minimum Detectable Group-Size Effects and Candidate Sample Sizes

# Sequence Sequence (S)	# Sessions	# Subjects	Minimum Detectable Effects	
			$\rho = 0.05$	$\rho = 0.10$
1	4	56	34.5 *pp*	40.6 *pp*
2	8	112	17.2 *pp*	20.3 *pp*
3	12	168	13.3 *pp*	15.7 *pp*
4	16	224	11.3 *pp*	13.3 *pp*

Note: *pp* stands for percentage point. Calculations use $\sigma = 0.25$ and 80% power.

Because the standard deviation of the subject-level mean probability is not known prior to the main data collection, I additionally report sensitivity of the MDEs to $\sigma \in \{0.20, 0.25, 0.30\}$ for the two most relevant candidate sample sizes. Table C2.4 shows that the gain from a fourth sequence block persists across the range of variance and ICC assumptions.

Table C2.4: Sensitivity of Minimum Detectable Effects to the Assumed Standard Deviation

σ	3 Sequences: 168 Subjects		4 Sequences: 224 Subjects	
	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.05$	$\rho = 0.10$
0.20	10.6 *pp*	12.6 *pp*	9.0 *pp*	10.6 *pp*
0.25	13.3 *pp*	15.7 *pp*	11.3 *pp*	13.3 *pp*
0.30	16.0 *pp*	18.8 *pp*	13.6 *pp*	16.0 *pp*

Note: MDEs are calculated at 80% power with a one-sided significance level of 5%.

Based on the pilot sessions conducted in April, the estimated average payment is around \$22 per subject, implying an expected total participant payment of approximately \$3,700. With the available experimental budget in the range of \$3,000 - \$4,000, I therefore target three complete sequence blocks. This sample size also provides reasonable statistical preci-

sion for the power-driving group-size comparison. Under the baseline assumptions $\sigma = 0.25$ and $\rho = 0.05$, the three-block design has 80% power to detect a group-size effect of approximately 13.3 percentage points. Under the more conservative ICC assumption $\rho = 0.10$, the corresponding minimum detectable effect is approximately 15.7 percentage points. The sensitivity analysis shows that the resulting MDEs remain within a comparable range under alternative assumptions about the subject-level standard deviation.